BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Assessment of malnutrition in adult patients with tuberculosis: a review of current practice

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-049777
Article Type:	Original research
Date Submitted by the Author:	04-Feb-2021
Complete List of Authors:	Ter Beek, Lies ; University Medical Centre Groningen, Pulmonary diseases and Tuberculosis Bolhuis, Mathieu; University Medical Centre Groningen, Clinical Pharmacy and Pharmacology Jager-Wittenaar, Harriët ; Hanzehogeschool Groningen, Brijan, Rene; UMCG, Pulmonary diseases and Tuberculosis Sturkenboom, Marieke; University Medical Centre Groningen, Clincal Pharmacy and Pharmacology Kerstjens, Huib; University Medical Centre Groningen, Pulmonology de Lange, Wiel; UMCG, Pulmonary diseases and Tuberculosis; UMCG, TB center Beatrixoord Tiberi, Simon; Barts Health NHS Trust, Department of Infection; Queen Mary University of London, Blizard Institute van der Werf, Tjip S.; University Medical Centre Groningen, Department of Pulmonary Diseases & Tuberculosis Alffenaar, Jan-Willem; University of Syndney, Schoold of Pharmacy, Faculty of Medicine and Health; Westmead Hospital Akkerman, Onno W.; UMCG, Pulmonary Diseases and Tuberculosis; UMCG, TB center Beatrixoord
Keywords:	NUTRITION & DIETETICS, Nutrition < TROPICAL MEDICINE, Tuberculosis < INFECTIOUS DISEASES

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Assessment of malnutrition in adult patients with tuberculosis: a review of current practice

Lies ter Beek*[1,2,3], Mathieu S. Bolhuis*[4], Harriët Jager-Wittenaar[5,6], René X.D. Brijan[3], Marieke G.G. Sturkenboom[4], Huib A.M. Kerstjens[1], Wiel C.M. de Lange[1,2], Simon Tiberi[7,8], Tjip S. van der Werf[1,9], Jan-Willem C. Alffenaar[10,11,12], Onno W. Akkerman[1,2*]

1. University of Groningen, University Medical Center Groningen, Department of Pulmonary Diseases and Tuberculosis, Groningen, The Netherlands
2. University of Groningen, University Medical Center Groningen, Tuberculosis Center Beatrixoord, Haren, The Netherlands
3. Department of Health Sciences, Faculty of Science, VU University Amsterdam, and Amsterdam Public Health Research Institute, Amsterdam, the Netherlands
4. University of Groningen, University Medical Center Groningen, Department of Clinical Pharmacy and Pharmacology, Groningen, The Netherlands.
5. Hanze University of Applied Sciences, Research Group Healthy Ageing, Allied Health Care and Nursing, Groningen, The Netherlands
6. University of Groningen, University Medical Center Groningen, Department of Oral and Maxillofacial Surgery, Groningen, The Netherlands
7. Department of Infection, Barts Health NHS Trust, London, United Kingdom
8. Blizard Institute, Queen Mary University of London, London, United Kingdom
9. University of Groningen, University Medical Center Groningen, Department of Internal Medicine, Groningen, The Netherlands
10. University of Sydney, Faculty of Medicine and Health, School of Pharmacy, Sydney, Australia
11. Westmead Hospital, Sydney, Australia
12. Marie Bashir Institute for Infectious Diseases and Biosecurity, University of Sydney, Sydney, Australia

***both authors contributed equally to this manuscript/study**

Running head: review of assessment of malnutrition and TB

Word count summary: 193

For peer review only

Word count text: 2374

Number of references: 89

Number of tables: 4

Number of figures: 1

Key words for indexing: nutritional assessment, undernutrition

***corresponding author**

Onno W. Akkerman, MD, PhD

Tuberculosis Center Beatrixoord

University Medical Center Groningen, PO Box 30002, 9750 RA Haren, The Netherlands

e-mail: o.w.akkerman@umcg.nl

phone: +31 50 3610857

fax: +31 50 3618152

For peer review only

ABSTRACT

Objectives Malnutrition is associated with a twofold risk of dying in patients with tuberculosis (TB) and considered an important potentially reversible risk factor for failure of TB treatment. The construct of malnutrition has three domains: intake or uptake of nutrition; body composition; and physical and cognitive function. The objectives of this review are to identify malnutrition assessment methods, and to establish the percentage of these assessment methods that cover the three domains of the construct of malnutrition, according to its international consensus definition.

Design Review of studies in patients with TB that reported assessment of malnutrition.

Results Seventeen assessment methods were identified in 69 included studies. In 53/69 (77%) of studies, body mass index was used as the only assessment method. Three out of 69 studies (4%) used a method that addressed all three domains of malnutrition.

Conclusions Most studies in adult TB patients did not describe their assessment method for malnutrition. Most TB patients are assessed for only one or two aspects of the conceptual definition of malnutrition. The use of international consensus criteria is recommended to establish systematic and uniform diagnostics and treatment of malnutrition.

Trials registration: PROSPERO with number CRD42019122832

ARTICLE SUMMARY

Strengths and limitations of this study

- PUBMED, CINAHL, and EMBASE were systematically searched for studies in the last decade, to prevent older studies describing outdated supportive care strategies which are no longer current practice to be eligible for inclusion.
- As there is no instrument for 'risk of bias assessment' of studies on diagnosis (e.g. malnutrition assessment), we evaluated risk of bias by scoring the presence of essential components required for adequate assessment and reporting of malnutrition.
- To report on how malnutrition was assessed in studies on TB, we determined the extent of coverage of the three domains of the conceptual definition of malnutrition.
- Implementation of guidelines in clinical practice and in study protocols takes time. This implementation might take longer than the time period used in this review.

For peer review only

Background

In 2019, tuberculosis (TB) was the worldwide leading cause of death from a single infectious agent, with 10 million new patients, and 1.2 million deaths [1]. TB prevalence is up to 20 times higher among people living in low-income countries compared to high-income countries [2]. TB, poverty and malnutrition are closely linked. Prevalence of malnutrition in patients with TB is estimated at 70%, and in this population, malnutrition is associated with a twofold risk of dying [3].

As hunger-related malnutrition caused by food-insecurity impacts the immune system [1, 4, 5], malnutrition is the most important risk factor for re-activation of TB, with a reported 27% attributable risk [6]. Malnutrition is considered an important, potentially reversible risk factor for treatment failure [7]. Therefore, a better understanding in assessing malnutrition in TB patients is urgently needed [8].

The European Society for Clinical Nutrition and Metabolism (ESPEN) defines malnutrition as 'a state resulting from lack of intake or uptake of nutrition that leads to altered body composition (decreased fat-free mass) and body cell mass leading to diminished physical and mental function and impaired clinical outcome from disease' [4]. In patients with TB, two types of malnutrition may co-exist: malnutrition without disease [4]; and disease-related malnutrition once active disease has developed. The latter is often driven by a combination of loss of appetite, malabsorption, and/or inflammation-driven catabolism [4, 5]. A low body mass index (BMI) is a characteristic for chronic malnutrition [9]. However, since malnutrition leads to loss of fat-free mass in all individuals, including those who are overweight or obese, patients with either a normal or high BMI may be malnourished as well [4]. In 2015, ESPEN published their first consensus on diagnostic criteria for malnutrition [9], followed by the Global Leadership Initiative on Malnutrition (GLIM) criteria in 2018, which were established by ESPEN, the American Society for Parenteral and Enteral Nutrition (ASPEN), la Federación Latino Americano de Terapia Nutriconal, Nutricion Clinica y Metabolismo (FELANPE), and the Parenteral and Enteral Nutrition Society of Asia (PENSA) [10].

It was not until 2013 that the World Health Organization (WHO) presented their first guideline on nutritional care and support for TB patients. In this guideline, the WHO stressed that all patients with active TB receive individualized nutritional assessment and management, including dietary counselling and nutritional interventions, to improve nutritional status and consequently, prevent TB treatment failure [11]. Nutritional assessment is a necessary step in the 'nutritional care process', enabling the professional to design a treatment plan together with the patient [12]. It is currently unknown to what extent the WHO guideline has been implemented in relation to diagnosing malnutrition TB patients.

For peer review only

The primary aim of this review is to identify malnutrition assessment methods that are used in adult TB patients. The secondary aim is to quantify the malnutrition assessment methods that reflect the international consensus definition for malnutrition in this population.

Methods

The protocol for this review was registered at PROSPERO with number CRD42019122832. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement was used [13].

Search strategy

In September 2020, PUBMED, CINAHL, and EMBASE were systematically searched for studies in any language. The search term consisted of a domain describing 'malnutrition' and a domain describing 'tuberculosis', whilst excluding *in vitro*, animal, and paediatric studies. Details of the search strategy can be found in the Supplementary table 1.

Inclusion and exclusion criteria

Studies in the English language that aimed to assess malnutrition and described a method for assessment of malnutrition in patients with microbiologically confirmed TB, published between January 1st, 2009 until September 18th, 2020 were considered eligible for inclusion. Since the aim of the review is to identify how malnutrition is assessed in current practice, 2009 was chosen as a starting point. Restricting the search to the last decade will prevent older studies describing outdated supportive care strategies which are no longer current practice to be eligible for inclusion. In addition, this time frame enables to compare 5 years before the publication of the WHO guideline on nutrition in patients with TB and 5 years after.

Only studies that focused on adult TB patients were included, because the assessment of malnutrition in children requires different methods than in adults as children are growing, and the criteria for measuring their nutritional status differ per by age [14].

Reviews and study protocols were excluded since they do not present original data. Case reports and abstracts/posters were also excluded since the information provided on methods used in this type of publication is considered too limited

Study selection

Screening of title and abstract was done independently by two authors using the Rayyan web application [15]. Final inclusion was based on an independent judgement of the full text of both

For peer review only

authors. Disagreements about inclusion were resolved through discussion and if consensus was not reached, a third author was consulted, which resulted in consensus in all cases.

Patient and public involvement

No patient involved

Data collection

Data collection was performed by two authors independently. The following characteristics were extracted from each study: citation, first author, country, year of publication, number of included patients, number of HIV co-infected patients, disease location, drug susceptibility of the *M. tuberculosis* isolate, and study design.

Data analysis

To report on how malnutrition was assessed in studies on TB, we determined the extent of coverage of the three domains of malnutrition, malnutrition included in the ESPEN conceptual definition of malnutrition, i.e., intake or uptake of nutrition (Domain A), body composition (Domain B), and physical and cognitive function (Domain C)[4, 16, 17].

Domain A was considered to be covered to some extent (+) if the method addressed nutritional intake or uptake at all. It was considered to be covered extensively (++) if the method addressed nutritional intake or uptake in depth. For domain B, weight change, BMI, and anthropometric measurements, such as skinfold measurements were considered covering domain B to some extent (+). It was considered extensively addressed (++) if the method included identification of muscle mass, lean mass, or fat-free mass. Domain C was considered to be covered to some extent (+) if functionality was addressed. Domain C was considered extensively covered (++) if physical, mental and cognitive function tests, or questions about activities of daily living were addressed.

As micronutrient or trace elements in serum are not representative for intake or uptake of protein or energy [4], laboratory tests were not considered to attribute to any of the malnutrition domains. Serum albumin and CRP are parameters for inflammation but are not related to parameters of protein/energy intake or functionality, nor do they represent body composition and were therefore not taken into consideration. In addition, while prealbumin (i.e. transthyretin) is sensitive for changes in protein and energy intake, this marker is influenced by inflammation activity [4].

As there is no instrument for 'risk of bias assessment' of studies on diagnosis (e.g. malnutrition assessment), we evaluated risk of bias by scoring the presence of essential components required for adequate assessment and reporting of malnutrition. The risk of bias in included studies was assessed by rating the following four characteristics:

For peer review only

- rationale for the assessment of malnutrition

-malnutrition was assessed and clear cut-off points were described in the method section. For example: 'malnutrition was assessed as BMI < 18.5 kg/m^2

-malnutrition was reported in the results section.

-the results with regard to malnutrition were reflected upon in the discussion section.

The characteristics were graded for quality of the study. Risk of bias was rated by - (meaning high risk of bias), + (medium risk of bias), or ++ (low risk of bias). The scores were added and the total was translated into very low (≥7 plusses), low (5-6 plusses), medium (3-4 plusses) or high risk of bias (≤2 plusses).

Results

The search resulted in 2175 studies after removal of duplicates. After screening by title and abstract, 272 studies were selected for a full text eligibility check. Review of the full text resulted in inclusion of 69 studies. A flow diagram of the selection process is visualized in **Figure 1**.

In total, among 69 studies, 17 different methods were used to assess malnutrition. Four studies used multiple (i.e., two to four) methods to assess malnutrition. Four studies used multiple criteria but integrated these together into one method to perform an assessment. In 53/69 (77%) of the studies, BMI was used as single assessment method. Among these studies eight different cut-off points were used, 34/53 studies (64%) used BMI < 18.5 kg/m^2. In seven studies, no cut-off values for BMI were described. **Table 1** shows all the cut-off values of BMI used as single method.

Coverage of the domains of malnutrition

Table 2 presents the coverage of the domains of malnutrition per assessment method. Uptake or intake of nutrition (A) was addressed to some extent by four methods that were used in five of the 69 studies. Body composition (B) was addressed to some extent by all but two studies, 67/69 (97%). Two methods did not address domain B, as these were methods that used self-reporting of diet quality and a food frequency questionnaire as assessment methods. Physical and/or cognitive functionality (C) was addressed to some extent by only 3/69 studies (4%).

Only two of the 17 different assessment methods that were used in the 69 studies covered all three domains of the definition, i.e., the Subjective Global Assessment (SGA) was used in two studies and the Mini Nutritional Assessment (MNA) was used in one study.

A descriptive subanalysis, as shown in **Table 3**, was performed to compare studies regarding their use of assessment tools that were published before and after the publication of the WHO guideline on nutritional care, and before and after the publication of the consensus criteria, respectively. No differences were found before publication of the WHO guideline versus after its publication,

For peer review only

regarding the use of BMI or methods that cover all three domains of the ESPEN definition of malnutrition.

Risk of bias assessment

Table 4 shows the risk of bias assessment and details of the 69 included studies. Twenty-one of the 69 studies (30%) did not describe the rationale for the assessment of malnutrition. Fifteen of the 69 studies (22%) described their assessment method without clear cut-off values. Risk of bias was very low in 24 studies (35%), low in 20 studies (29%), medium in 20 studies (29%) and high in five studies (7%).

Discussion

This review showed that most studies in adult TB patients that included malnutrition as a variable did not describe their assessment method. Most TB patients are assessed for only one or two aspects of the construct of malnutrition. BMI is often used assessment method for malnutrition in studies with TB patients, even though it only partly covers just one malnutrition domain (domain B). In addition, while some studies did not report a cut-off value for BMI, in other studies different cut-off values for BMI were used, therefore making it difficult to compare these studies. The use of BMI could be justified from a public health perspective, since a low BMI is a characteristic of chronic malnutrition that involves loss of both fat and muscle tissue[9]. However, the use of BMI alone for assessing malnutrition in this population is debatable. In clinical settings, disease-related malnutrition is the predominant type of malnutrition. Disease-related malnutrition is a (sub)acute condition, in which loss of weight and muscle/fat-free mass does not automatically result in a low BMI, while loss of weight and fat-free mass are related to poor clinical outcomes including increased morbidity and mortality [9, 18]. With the current global overweight and obesity epidemic, patients with catabolic diseases such as TB may lose more than 20% of their weight and muscle mass within three to six months, and still show BMI values at or above normal range [19]. When assessing malnutrition in TB patients solely based on BMI, these patients will not be identified when they are malnourished, despite the therapeutic and prognostic implications of malnutrition. [20]

This review indicates that the international criteria for the assessment of malnutrition have not yet found their way into studies with TB patients. BMI is by far the most frequently used assessment method for malnutrition (77%) in studies with TB patients. Only a few (4%) of the studies that reported on the assessment methods addressed all domains of the conceptual definition of malnutrition. For the studies performed after the ESPEN criteria of 2015 [21] and the GLIM criteria of 2018 [10], this may be explained by the fact that the WHO does not yet refer to these criteria in their communications. The GLIM criteria have been developed for global use and is therefore

For peer review only

recommended to be used in any setting for all TB patients. The GLIM criteria do not include the domain of functionality as a criteria.

There are some limitations in our study. First, our study focussed on published articles and not on the underlying study protocol. In some cases, detailed information on the assessment and operationalisation of malnutrition might be available in the study protocol, however it was not addressed in the article making it unavailable to the public domain. Second, implementation of guidelines in clinical practice and in study protocols takes time. This implementation might take longer than the time period used in this review. Third, we used the publication date in relation to the publication of the WHO guideline. The time between data-collection and publication, might cover the moment of publication of the WHO guideline.

Awareness of the presence of malnutrition and concept of malnutrition assessment in health care professionals working with TB patients needs improvement. Future studies regarding malnutrition assessment in TB patients should aim at implementation of international consensus guidelines regarding malnutrition assessment. Using the same terminology for malnutrition may make a difference to our patients, improve outcomes as well as reduce chronic sequelae and help to end TB.

In conclusion, most studies in adult TB patients did not describe their assessment method for malnutrition. Most TB patients are assessed for only one or two aspects of the conceptual definition of malnutrition. Various methods for assessment of malnutrition have been used, and only a very small proportion of the published studies on TB used an assessment method that fully reflects the definition of malnutrition. The use of international consensus criteria is recommended to establish systematic and uniform diagnostics and treatment of malnutrition.

For peer review only

Funding statement

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing Interests statement

H. Jager-Wittenaar was co-developer of the Patient-Generated Subjective Global Assessment©-based Pt-Global© app/web tool.

Other authors have no conflict of interest to disclose.

Author contributions

Lies ter Beek: conception and design, data collection, acquisition, analysis, interpretation of data; drafting

Mathieu S. Bolhuis: conception and design, data collection, acquisition, analysis, interpretation of data; revising draft critically for important intellectual content

Harriët Jager-Wittenaar: interpretation of data and revising draft critically for important intellectual content

René X.D. Brijan: acquisition, analysis, interpretation of data; drafting

Marieke G.G. Sturkenboom: interpretation of data, and revising draft critically for important intellectual content

Huib A.M. Kerstjens: interpretation of data, and revising draft critically for important intellectual content

Wiel C.M. de Lange: interpretation of data, and revising draft critically for important intellectual content

Simon Tiberi: interpretation of data, and revising draft critically for important intellectual content

Tjip S. van der Werf: interpretation of data, and revising draft critically for important intellectual content

Jan-Willem C. Alffenaar: interpretation of data, and revising draft critically for important intellectual content

Onno W. Akkerman: conception and design, interpretation of data; revising draft critically for important intellectual content

Data availability

No additional data available

Acknowledgements

For peer review only

We thank Martine Sealy for her contribution to the conceptualization and visualization of this review. We thank Giovanni Sotgiu for his feedback on the manuscript. We thank the Central Medical Library of the University of Groningen for their help with designing our search strategy.

Table 1 Cut-off values BMI exclusively used

Cut-off value	Studies using only BMI for assessment of malnutrition n= 53
≤ 20.0 kg/m^2	1
< 20.0 kg/m^2	2
< 18.5 kg/m^2	34
≤ 18.5 kg/m^2	3
< 18.49 kg/m^2	1
< 18.4 kg/m^2	1
< 17.0 kg/m^2	2
< 16.0 kg/m^2	2
No clear cut-off value described in study	7

For peer review only

Table 2 Coverage of the domains of malnutrition by 17 different assessment methods

DOMAINS **Description of domain**	**A** **intake or uptake of nutrition**	**B** **body composition**	**C** **physical and cognitive function**	**Total**
PIBW, BMI, Albumin, TLC, Cholesterol, Hb	-	+	-	+
Weight change	-	+	-	+
BMI & MUAC	-	+	-	+
Self-report of diet quality	+	-	-	+
MNA	++	+	++	+++++
MUST	+	+	-	++
AMA	-	++	-	++
BMI/Albumin	-	+	-	+
BMI, MUAC, TSF, MAMC, Hb, Ht, albumin, total protein, globulin, iron fixation capacity, retinol, tocopherol Zn, Se, Fe	-	+	-	+
MAMC	-	++	-	++
underweight	-	+	-	+
TSF/TST	-	+	-	+
SGA	++	++	++	++++++
BMI	-	+	-	+
BIA: (Body fat%)	-	+	-	+
MUAC	-	+	-	+
FFQ	++	-	-	++

Abbreviations: *PIBW: Percent Ideal Body Weight; BMI: Body Mass index; TLC: Total Lymphocyte Count; Hb: Haemoglobin; Ht: Haematocrit; MUAC: Mid Upper Arm Circumference; MNA: Mini Nutritional Assessment; AMA: Arm Muscle Area; MAMC: Mid Arm Muscle Circumference; Zn: Zinc; Se: Selenium; Fe: Iron; TSF: Triceps Skin Fold ; TST: Triceps Skinfold Thickness; SGA: Subjective Global Assessment; BIA: Bio-electrical Impedance Analysis; FFQ: Food Frequency Questionnaire*

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3 Subanalysis of malnutrition assessment methods across the recent decade

	Number of studies per year (n=69)	BMI as the only assessment method		Use of an assessment method that attributes to 3 domains	
		N	%		%
2009	3	3	(100)	-	
2010	7	5	(71)	-	
2011	4	3	(75)	1	(25)
2012	1	1	(100)	-	
2013 **Publication of WHO guideline nutritional care in TB**	8	4	(50)	1	(13)
2014	4	2	(50)	-	
2015 **Publication of ESPEN consensus diagnostic criteria for assessment of malnutrition**	5	4	(80)	-	
2016	5	5	(100)	-	
2017	3	2	(67)	-	
2018 **Publication of GLIM consensus diagnostic criteria for assessment of malnutrition**	6	5	(83)	1	(17)
2019	17	14	(82)	-	
2020	6	5	(83)	-	
Total	**69**	**53**		**3**	

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 4. Details of included studies and risk of bias assessment

First author	Year	Country	Rationale	Assessment method	Clear cut-off	Maln * in results	Maln * in discussion	Risk of bias	Number of patients	HIV coinfection	Susceptibility	Type
Martins[22]	2009	Timor-Leste	+	BMI < 18.5 kg/m²	-	+	+	medium	270	unknown	unknown	pulmonary
Pakasi[23]	2009	Indonesia	++	BMI < 18.5 kg/m²	++	++	++	very low	121	unknown	unknown	pulmonary
Ulasli[24]	2009	Turkey	+	BMI < 20.0 kg/m²	++	+	+	low	24	unknown	unknown	pulmonary and/or extra pulmonary
Kim[25]	2010	South-Korea	++	PIBW, BMI, Albumin, TLC, Cholesterol, Hb	++	++	++	very low	23	unknown	unknown	pulmonary
Khoharo[26]	2010	Pakistan	-	BMI < 18.5 kg/m²	++	+	++	low	350	unknown	unknown	pulmonary
Pakasi[27]	2010	Indonesia	+	BMI (no clear cut-off)	-	+	+	medium	300	unknown	unknown	pulmonary
Gambhir[28]	2010	India	-	BMI <18.5 kg/m²	++	+	+	medium	95	no	unknown	pulmonary and/or extra pulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Singla[29]	2010	India	+	BMI < 18.5 kg/m²	+	+	+	medium	175	no	unknown	pulmonary and/or extra pulmonary
Mupere[30]	2010	Uganda	++	BMI < 18.5 kg/m²	+	+	+	low	445	115	unknown	pulmonary
Warmelink[31]	2010	Nederland	+	Change in body weight	-	+	+	medium	192	15	MDR patients included	pulmonary and/or extra pulmonary
Podewils[32]	2011	Latvia	++	BMI < 18.5 kg/m²	++	++	+	very low	995	32	only MDR patients	pulmonary
De Jong[33]	2011	Gambia	-	BMI < 16 kg/m²	++	+	+	medium	692	56	unknown	pulmonary
Kawai[34]	2011	Tanzania	+	BMI < 18.5 kg/m²	++	+	+	low	887	471	unknown	pulmonary
Miyata[35]	2011	Japan	++	SGA	++	++	++	very low	39	unknown	unknown	pulmonary
Mupere[36]	2012	Uganda	-	BMI < 18.5 kg/m²	++	+	+	medium	747	539	unknown	pulmonary
Piva[37]	2013	Brazil	++	BMI < 18.5 kg/m²	++	++	++	very low	34	unknown	unknown	pulmonary
Islam[38]	2013	Bangladesh	++	BMI < 18.5 kg/m² and MUAC < 22 cm	++	+	++	very low	1068	unknown	unknown	pulmonary

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Chittoor[39]	2013	Mexico	-	Self-reported estimate of diet quality	-	+	+	high	75	unknown	unknown	pulmonary
Miyata[40]	2013	Japan	+	MNA < 17	++	+	+	low	53	unknown	unknown	pulmonary
Bhargava[3]	2013	India	++	BMI < 18.5 kg/m^2	++	++	++	very low	1695	39	unknown	pulmonary
Bakari[41]	2013	Tanzania	++	BMI < 18.5 kg/m^2	++	+	+	low	43	43	unknown	pulmonary
Ismawati[42]	2013	Indonesia	+	BMI (no clear cut-off)	-	+	+	medium	30	unknown	unknown	pulmonary
Miyata[43]	2013	Japan	++	MUST	++	+	+	low	57	unknown	unknown	pulmonary
Maeda[44]	2014	Vietnam	-	BMI ≤ 18.5 kg/m^2	++	+	+	medium	465	38	Including different types of resistance	pulmonary
Oliveira[45]	2014	Brazil	++	TST AMA	+	++	++	very low	166	31	unknown	pulmonary
Tian[46]	2014	China	-	BMI < 18.5 kg/m^2 and/or serum albumin < 30 g/L	++	+	+	medium	160	43	unknown	pulmonary
Kumar[47]	2014	India	+	BMI (no clear cut-off)	-	+	+	medium	376	unknown	Only MDR	pulmonary
Bacelo[48]	2015	Brazil	++	Multiple anthropometric and biochemical biomarkers	++	++	++	very low	68	22	unknown	Pulmonary and extra pulmonary
Golemba[49]	2015	Argentina	-	BMI ≤ 20 kg/m^2	++	+	+	medium	75	0	unknown	pulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Te Brake[50]	2015	Indonesia	++	BMI < 18.5 kg/m²	++	+	++	very low	36	0	unknown	pulmonary
Medellin-Garibay[51]	2015	Mexico	-	BMI < 18.49 kg/m²	+	+	+	medium	48	0	unknown	Pulmonary and extrapulmonary
Ezeamama[52]	2015	Uganda	++	BMI < 18.5 kg/m²	++	++	++	very low	208	208	unknown	pulmonary
McLachlan[53]	2016	South Africa	-	BMI < 18.5 kg/m²	++	-	+	medium	105	74	Susceptible and MDR	Pulmonary and extrapulmonary
Gebrecherkos[54]	2016	Ethiopia	+	BMI < 18.5 kg/m²	++	+	+	Low	15	4	No rifampicin resistance	pulmonary
Araújo-Mariz[55]	2016	Brazil	-	BMI < 18.5 kg/m²	++	+	+	medium	173	173	unknown	unknown
Buntoro[56]	2016	Indonesia	+	BMI <18.5 kg/m²	++	++	++	very low	72	unknown	unknown	pulmonary
Pandit[57]	2016	India	-	BMI <20.0 kg/m²	-	+	+	high	148	unknown	unknown	pulmonary
Abdelbary[58]	2017	Mexico	-	Underweight	-	++	-	high	8431	447	Susceptible and MDR	pulmonary and extrapulmonary
Bhat[59]	2017	India	+	BMI < 18.5 kg/m²	++	+	+	low	267	0	unknown	pulmonary
Hochberg[60]	2017	India	-	BMI < 18.5 kg/m²	++	++	++	low	409	1	Only normal susceptible	pulmonary
Piparva[61]	2018	India	-	BMI < 18.5 kg/m²	++	++	++	low	108	5	only MDR	pulmonary
Gurung[62]	2018	Nepal	++	BMI (no clear cut-off)	-	++	++	low	133	1	unknown	pulmonary and extrapulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Pande[63]	2018	India	-	BMI < 18.4 kg/m^2	+	+	+	medium	728	53	unknown	pulmonary and extrapulmonary
Rao[64]	2018	India	-	BMI (no clear cut-off)	-	+	++	medium	220	unknown	unknown	pulmonary
Sattler[65]	2018	4 continents, 26 study sites	+	BMI < 18.5 kg/m^2	-	+	+	medium	51	51	unknown	pulmonary
Kirchmann Lazzari[66]	2018	Brazil	++	BMI TSF MAMC SGA	+	++	++	very low	108	44	unknown	pulmonary
Cheng[67]	2019	China	-	BMI < 18 kg/m^2	+	+	-	high	85	0	unknown	intestinal
Cavalheiro Skupien[68]	2019	Brazil	-	BMI ≤ 18.5 kg/m^2	-	+	+	high	35	unknown	unknown	pulmonary
Abdullahi[69]	2019	Kenya	+	BMI < 18.5 kg/m^2	++	++	-	low	10717	3163	unknown	Pulmonary and extrapulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Benzekri[70]	2019	Senegal	-	BMI (no clear cut-off)	-	++	+	medium	26	26	Susceptible	pulmonary
Chebrolu[71]	2019	USA	++	BMI < 16 kg/m²	++	+	+	low	27	0	unknown	Pulmonary and extrapulmonary
Da Silva[72]	2019	Brazil	+	BMI < 18.5 kg/m² TSF MUAC MAMC BIA FFQ	+	++	++	low	35	14	unknown	Pulmonary
Feleke[73]	2019	Ethiopia	++	BMI < 18.5 kg/m²	++	++	+	very low	1681	595	unknown	Pulmonary and extrapulmonary
Gashaw[74]	2019	Ethiopia	++	BMI ≤ 18.5 kg/m² MUAC ≤ 23 cm (men) MUAC ≤ 22 cm (women)	++	++	++	very low	384	unknown	unknown	Pulmonary and extrapulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Hoyt[75]	2019	India	++	BMI < 18.5 kg/m²	++	++	++	very low	173	unknown	unknown	pulmonary
Hussien[76]	2019	Ethiopia	++	BMI < 18.5 kg/m²	+	++	++	very low	372	42	unknown	pulmonary
Lee[77]	2019	Philippines	++	BMI< 17 kg/m² MUAC ≤ 20.5 cm (men) MUAC ≤ 18.5 cm (women)	++	++	++	very low	348	22	31 MDR	unknown
Mailu[78]	2019	Kenya	-	BMI < 18.5 kg/m²	++	++	+	low	421409	134776	unknown	Pulmonary and extrapulmonary
Rashak[79]	2019	Mexico	++	BMI < 18.5 kg/m² (or BMI ≤ 18.5 kg/m², not clearly described)	-	++	++	low	5508	224	unknown	Pulmonary and extrapulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Ren[80]	2019	China	++	BMI < 18.5 kg/m^2	++	-	++	low	300	unknown	unknown	pulmonary
Wardani[81]	2019	Indonesia	+	BMI ≤ 18.5 kg/m^2	++	+	-	medium	311	unknown	unknown	pulmonary
Wessels[82]	2019	South Africa	++	BMI < 18.5 kg/m^2 (underweight)	-	++	+	low	100	68	unknown	unknown
White[83]	2019	Philippines	++	BMI < 17 kg/m^2	++	++	++	very low	348	22	31	unknown
Ma'rufi[84]	2020	Indonesia	++	BMI < 18.5 kg/m^2	++	++	++	very low	200	0 (excluded)	0 (excluded)	pulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Musuenge[85]	2020	Burkino Faso	++	BMI < 18.5 kg/m^2	++	++	++	Very low	302	23	unknown	pulmonary
Seid[86]	2020	Ethiopia	++	BMI < 18.5 kg/m^2	+	++	++	Very low	284	51	Unknown	Pulmonary and extrapulmonary
Edwards[87]	2020	Phillipines	+	BMI < 17.0 kg/m^2	+	++	++	Very low	446	68 (28%)	Susceptible	Unknown
White[88]	2020	Phillipines	++	BMI < 18.5 kg/m^2	+	++	++	Very low	637	74 (24%)	Unknown	Unknown
Mollah[89]	2020	India	++	BMI < 18.5 kg/m^2 MUAC <23 (m) & <22 (f)	++	-	++	Low	113	Unknown	Unknown	Pulmonary and extrapulmonary

Abbreviations *PIBW: Percent Ideal Body Weight; BMI: Body Mass index; TLC: Total Lymphocyte Count; Hb: Haemoglobin; MUAC: Mid Upper Arm Circumference; MNA: Mini Nutritional Assessment;*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

AMA: Arm Muscle Area; MAMC: Mid Arm Muscle Circumference; TSF: Triceps Skin Fold ; TST: Triceps Skinfold Thickness; SGA: Subjective Global Assessment; BIA: Bio-electrical Impedance Analysis; FFQ: Food Frequency Questionnaire

For peer review only

References

1 World Health Organisation (WHO). Global Tuberculosis Report 2020 2020;Geneva, WHO.

2 World Health Organization (WHO). Addressing poverty in TB control - options for national TB control programmes 2005;Geneva, WHO.

3 Bhargava A, Chatterjee M, Jain Y, et al. Nutritional status of adult patients with pulmonary tuberculosis in rural central India and its association with mortality. *PLoS One* 2013;8:e77979 doi:10.1371/journal.pone.0077979 [doi].

4 Cederholm T, Barazzoni R, Austin P, et al. ESPEN guidelines on definitions and terminology of clinical nutrition. *Clin Nutr* 2017;36:49-64 doi:S0261-5614(16)31242-0 [pii].

5 Sobotka L, Allison SP, Forbes A, et al. Basics in clinical nutrition. Prague, Czech Republic: Publishing House Galen 2011.

6 Dheda K, Barry CE,3rd, Maartens G. Tuberculosis. *Lancet* 2016;387:1211-26 doi:10.1016/S0140-6736(15)00151-8 [doi].

7 Choi R, Jeong BH, Koh WJ, et al. Recommendations for Optimizing Tuberculosis Treatment: Therapeutic Drug Monitoring, Pharmacogenetics, and Nutritional Status Considerations. *Ann Lab Med* 2017;37:97-107 doi:10.3343/alm.2017.37.2.97 [doi].

8 World Health Organization (WHO). The End TB Strategy 2014;Geneva, WHO.

9 Cederholm T, Bosaeus I, Barazzoni R, et al. Diagnostic criteria for malnutrition - An ESPEN Consensus Statement. *Clin Nutr* 2015;34:335-40 doi:S0261-5614(15)00075-8 [pii].

10 Cederholm T, Jensen GL, Correia MITD, et al. GLIM criteria for the diagnosis of malnutrition - A consensus report from the global clinical nutrition community. *J Cachexia Sarcopenia Muscle* 2019;10:207-17 doi:10.1002/jcsm.12383 [doi].

11 World Health Organisation (WHO). Nutritional care and support for patients with tuberculosis 2013;Geneva, WHO doi:NBK189867 [bookaccession].

12 Lacey K, Pritchett E. Nutrition Care Process and Model: ADA adopts road map to quality care and outcomes management. *J Am Diet Assoc* 2003;103:1061-72 doi:10.1053/jada.2003.50564 [doi].

13 Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;6:e1000097 doi:10.1371/journal.pmed.1000097 [doi].

14 Turck D, Braegger CP, Colombo C, et al. ESPEN-ESPGHAN-ECFS guidelines on nutrition care for infants, children, and adults with cystic fibrosis. *Clin Nutr* 2016;35:557-77 doi:10.1016/j.clnu.2016.03.004 [doi].

15 Ouzzani M, Hammady H, Fedorowicz Z, et al. Rayyan-a web and mobile app for systematic reviews. *Syst Rev* 2016;5:210,016-0384-4 doi:10.1186/s13643-016-0384-4 [doi].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16 Ter Beek L, Vanhauwaert E, Slinde F, et al. Unsatisfactory knowledge and use of terminology regarding malnutrition, starvation, cachexia and sarcopenia among dietitians. *Clin Nutr* 2016;35:1450-6 doi:S0261-5614(16)30012-7 [pii].

17 Sealy MJ, Nijholt W, Stuiver MM, et al. Content validity across methods of malnutrition assessment in patients with cancer is limited. *J Clin Epidemiol* 2016;76:125-36 doi:10.1016/j.jclinepi.2016.02.020 [doi].

18 Correia MI, Waitzberg DL. The impact of malnutrition on morbidity, mortality, length of hospital stay and costs evaluated through a multivariate model analysis. *Clin Nutr* 2003;22:235-9 doi:S0261561402002157 [pii].

19 Gonzalez MC, Correia MITD, Heymsfield SB. A requiem for BMI in the clinical setting. *Curr Opin Clin Nutr Metab Care* 2017;20:314-21 doi:10.1097/MCO.0000000000000395 [doi].

20 Jensen GL, Mirtallo J, Compher C, et al. Adult starvation and disease-related malnutrition: a proposal for etiology-based diagnosis in the clinical practice setting from the International Consensus Guideline Committee. *JPEN J Parenter Enteral Nutr* 2010;34:156-9 doi:10.1177/0148607110361910; 10.1177/0148607110361910.

21 Cederholm T, Bosaeus I, Barazzoni R, et al. Diagnostic criteria for malnutrition - An ESPEN Consensus Statement. *Clin Nutr* 2015;34:335-40 doi:10.1016/j.clnu.2015.03.001 [doi].

22 Martins N, Morris P, Kelly PM. Food incentives to improve completion of tuberculosis treatment: randomised controlled trial in Dili, Timor-Leste. *BMJ* 2009;339:b4248 doi:10.1136/bmj.b4248 [doi].

23 Pakasi TA, Karyadi E, Dolmans WM, et al. Malnutrition and socio-demographic factors associated with pulmonary tuberculosis in Timor and Rote Islands, Indonesia. *Int J Tuberc Lung Dis* 2009;13:755-9.

24 Ulasli SS, Ulubay G, Arslan NG, et al. Characteristics and Outcomes of End-Stage Renal Disease Patients With Active Tuberculosis Followed in Intensive Care Units. *Saudi J Kidney Dis Transpl* 2009;20:254-9.

25 Kim JS, Wilson JM, Lee SR. Dietary implications on mechanisms of sarcopenia: roles of protein, amino acids and antioxidants. *J Nutr Biochem* 2010;21:1-13 doi:10.1016/j.jnutbio.2009.06.014; 10.1016/j.jnutbio.2009.06.014.

26 Khoharo HK, Ansari S, Siddiqui AA, et al. Standard antituberculosis drug induced hepatotoxicity: Do the risk factors matter?. *J Liaquat Uni Med Health Sci* 2010;9:84-7.

27 Pakasi TA, Karyadi E, Suratih NMD, et al. Zinc and vitamin A supplementation fails to reduce sputum conversion time in severely malnourished pulmonary tuberculosis patients in Indonesia. *Nutr J* 2010;9:41.

28 Gambhir HS, Kaushik RM, Kaushik R, et al. Tobacco smoking-associated risk for tuberculosis: a case-control study. *Int Health* 2010;2:216-22.

29 Singla R, Sharma SK, Mohan A, et al. Evaluation of risk factors for antituberculosis treatment induced hepatotoxicity. *Indian J Med Res* 2010;132:81-6.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

30 Mupere E, Zalwango S, Chiunda A, et al. Body composition among HIV-seropositive and HIV-seronegative adult patients with pulmonary tuberculosis in Uganda. *Ann Epidemiol* 2010;20:210-6.

31 Warmelink I, van Altena R, ten Hacken N, et al. Nutritional status and vitamin D3 during antimicrobial treatment. *Lancet* 2011;377:1407-8.

32 Podewils LJ, Holtz T, Riekstina V, et al. Impact of malnutrition on clinical presentation, clinical course, and mortality in MDR-TB patients. *Epidemiol Infect* 2011;139:113-20 doi:10.1017/S0950268810000907 [doi].

33 de Jong BC, Adetifa I, Walther B, et al. Differences between tuberculosis cases infected with Mycobacterium africanum, West African type 2, relative to Euro-American Mycobacterium tuberculosis: an update. *FEMS Immunol Med Microbiol* 2010;58:102-5 doi:10.1111/j.1574-695X.2009.00628.x [doi].

34 Kawai K, Villamor E, Mugusi FM, et al. Predictors of change in nutritional and hemoglobin status among adults treated for tuberculosis in Tanzania. *Int J Tuberc Lung Dis* 2011;15:1380-9 doi:10.5588/ijtld.10.0784 [doi].

35 Miyata S, Tanaka M, Ihaku D. Subjective global assessment in patients with pulmonary tuberculosis. *Nutr Clin Pract* 2011;26:55-60 doi:10.1177/0884533610392380 [doi].

36 Mupere E, Malone L, Zalwango S, et al. Lean tissue mass wasting is associated with increased risk of mortality among women with pulmonary tuberculosis in urban Uganda. *Ann Epidemiol* 2012;22:466-73.

37 Piva SG, Costa Mda C, Barreto FR, et al. Prevalence of nutritional deficiency in patients with pulmonary tuberculosis. *J Bras Pneumol* 2013;39:476-83 doi:10.1590/S1806-37132013000400012 [doi].

38 Islam QS, Ahmed SM, Islam MA, et al. Beyond drugs: tuberculosis patients in Bangladesh need nutritional support during convalescence. *Public health action* 2013;3:136-40.

39 Chittoor G, Arya R, Farook VS, et al. Epidemiologic investigation of tuberculosis in a Mexican population from Chihuahua State, Mexico: a pilot study. *Tuberculosis (Edinburgh, Scotland)* 2013;93:S71-7.

40 Miyata S, Tanaka M, Ihaku D. Full mini nutritional assessment and prognosis in elderly patients with pulmonary tuberculosis. *J Am Coll Nutr* 2013;32:307-11 doi:10.1080/07315724.2013.826114 [doi].

41 Bakari M, Wamsele J, MacKenzie T, et al. Nutritional status of HIV-infected women with tuberculosis in Dar es Salaam, Tanzania. *Public health action* 2013;3:224-9.

42 Ismawati. R., Bambang W, Priatna DY, et al. The effect of zinc, lysine and vitamin a supplementation to increase cellular immune response of pulmonary tuberculosis patients. *J Trace Elem Med Biol* 2013;27:21.

43 Miyata S, Tanaka M, Ihaku D. The prognostic significance of nutritional status using malnutrition universal screening tool in patients with pulmonary tuberculosis. *Nutrition journal* 2013;12:42.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

44 Maeda S, Hang NT, Lien LT, et al. Mycobacterium tuberculosis strains spreading in Hanoi, Vietnam: Beijing sublineages, genotypes, drug susceptibility patterns, and host factors. *Tuberculosis (Edinb)* 2014;94:649-56 doi:S1472-9792(14)20533-6 [pii].

45 Oliveira MG, Delogo KN, Oliveira,Hedi Marinho de Melo Gomes de, et al. Anemia in hospitalized patients with pulmonary tuberculosis. *Jornal brasileiro de pneumologia : publicacao oficial da Sociedade Brasileira de Pneumologia e Tisilogia* 2014;40:403-10.

46 Tian PW, Wang Y, Shen YC, et al. Different risk factors of recurrent pulmonary tuberculosis between Tibetan and Han populations in Southwest China. *Eur Rev Med Pharmacol Sci* 2014;18:1482-6 doi:7406 [pii].

47 Kumar NSS, Hemraj SK, Dutt RA. Phase angle measurement in pulmonary tuberculosis patients and control subjects using bio-impedance analysis. *The Indian journal of tuberculosis* 2014;61:224-31.

48 Bacelo AC, Ramalho A, Brasil PE, et al. Nutritional Supplementation Is a Necessary Complement to Dietary Counseling among Tuberculosis and Tuberculosis-HIV Patients. *PLoS One* 2015;10:e0134785 doi:10.1371/journal.pone.0134785 [doi].

49 Golemba AS, Ferreyra FG, Martearena RE, et al. Drug-induced hepatotoxicity and tuberculosis in a hospital from the Argentinian northeast: cross-sectional study. *Medwave* 2015;15:e6135 doi:10.5867/medwave.2015.04.6135 [doi].

50 Te Brake LH, Ruslami R, Later-Nijland H, et al. Exposure to total and protein-unbound rifampicin is not affected by malnutrition in Indonesian tuberculosis patients. *Antimicrob Agents Chemother* 2015;74:986-90 doi:AAC.03485-14 [pii].

51 Medellin-Garibay SE, Cortez-Espinosa N, Milan-Segovia RC, et al. Clinical Pharmacokinetics of Rifampin in Patients with Tuberculosis and Type 2 Diabetes Mellitus: Association with Biochemical and Immunological Parameters. *Antimicrob Agents Chemother* 2015;59:7707-14 doi:10.1128/AAC.01067-15 [doi].

52 Ezeamama AE, Mupere E, Oloya J, et al. Age, sex, and nutritional status modify the CD4+ T-cell recovery rate in HIV-tuberculosis co-infected patients on combination antiretroviral therapy. *Int J Infect Dis* 2015;35:73-9 doi:10.1016/j.ijid.2015.04.008 [doi].

53 McLachlan I, Visser WI, Jordaan HF. Skin conditions in a South African tuberculosis hospital: Prevalence, description, and possible associations. *Int J Dermatol* 2016;55:1234-41.

54 Gebrecherkos T, Gelaw B, Tessema B. Smear positive pulmonary tuberculosis and HIV co-infection in prison settings of North Gondar Zone, Northwest Ethiopia. *BMC Public Health* 2016;16:1091.

55 Araújo-Mariz C, Lopes EP, Acioli-Santos B, et al. Hepatotoxicity during Treatment for Tuberculosis in People Living with HIV/AIDS. *PloS one* 2016;11:e0157725.

56 Buntoro IF, Kristin E, Sumardi. Decrease of liver function after treatment of antituberculosis drugs in tuberculosis patients with malnutrition and alcohol consumption. *Int J Pharm Pharm Sci* 2016;8:269-73.

57 Pandit A, Pandey Ak. Liver dysfunction in pulmonary tuberculosis patients on dots: A study and review. *J Gastroenterol Hepatol Res* 2016;5:2254-60.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

58 Abdelbary BE, Garcia-Viveros M, Ramirez-Oropesa H, et al. Predicting treatment failure, death and drug resistance using a computed risk score among newly diagnosed TB patients in Tamaulipas, Mexico. *Epidemiol Infect* 2017;145:3020-34.

59 Bhat J, Rao VG, Sharma RK, et al. Investigation of the risk factors for pulmonary tuberculosis: A case-control study among Saharia tribe in Gwalior district, Madhya Pradesh, India. *Indian J Med Res* 2017;146:97-104.

60 Hochberg NS, Sarkar S, Horsburgh CR, et al. Comorbidities in pulmonary tuberculosis cases in Puducherry and Tamil Nadu, India: Opportunities for intervention. *PloS one* 2017;12:e0183195.

61 Piparva KG, Jansari G, Singh AP. Evaluation of treatment outcome and adverse drug reaction of directly observed treatment (DOT) plus regimen in multidrug-resistant tuberculosis (MDR-TB) patients at district tuberculosis centre Rajkot. *Perspectives in clinical research* 2018;9:165-9.

62 Gurung LM, Bhatt LD, Karmacharya I, et al. Dietary Practice and Nutritional Status of Tuberculosis Patients in Pokhara: A Cross Sectional Study. *Frontiers in nutrition* 2018;5:1-6.

63 Pande T, Huddart S, Xavier W, et al. Prevalence of diabetes mellitus amongst hospitalized tuberculosis patients at an Indian tertiary care center: A descriptive analysis. *PloS one* 2018;13:e0200838.

64 Rao VG, Bhat J, Yadav R, et al. A comparative study of the socio-economic risk factors for pulmonary tuberculosis in the Saharia tribe of Madhya Pradesh, India. *Trans R Soc Trop Med Hyg* 2018;112:272-8.

65 Sattler FR, Chelliah D, Wu X, et al. Biomarkers Associated with Death After Initiating Treatment for Tuberculosis and HIV in Patients with Very Low CD4 Cells. *Pathogens & immunity* 2018;3:46-62.

66 Lazzari T.K., Forte G.C., Silva D.R. Nutrition Status Among HIV-Positive and HIV-Negative Inpatients with Pulmonary Tuberculosis. *Nutr.Clin.Prac.* 2018;33:858-64.

67 Cheng W, Zhang S, Li Y, et al. Intestinal tuberculosis: clinico-pathological profile and the importance of a high degree of suspicion. *Trop Med Int Health* 2019;24:81-90.

68 Skupien EC, Lazzari T, Coutinho SE, et al. The relation between leptin and inflammatory markers with respiratory and peripheral muscle strength in tuberculosis: A case-control study. *Clin Respir J* 2018;12:2559-65.

69 Abdullahi OA, Ngari MM, Sanga D, et al. Mortality during treatment for tuberculosis; a review of surveillance data in a rural county in Kenya. *PloS one* 2019;14:e0219191.

70 Benzekri NA, Sambou JF, Tamba IT, et al. Nutrition support for HIV-TB co-infected adults in Senegal, West Africa: A randomized pilot implementation study. *PloS one* 2019;14:e0219118.

71 Chebrolu P., Laux T., Chowdhury S., et al. The risk of refeeding syndrome among severely malnourished tuberculosis patients in Chhattisgarh, India. *Indian J Tuberc* 2019:https://doi.org/10.1016/j.ijtb.2019.03.004.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

72 da Silva LF, Skupien EC, Lazzari T, et al. Advanced glycation end products (AGE) and receptor for AGE (RAGE) in patients with active tuberculosis, and their relationship between food intake and nutritional status. *PloS one* 2019;14:e0213991.

73 Feleke BE, Feleke TE, Biadglegne F. Nutritional status of tuberculosis patients, a comparative cross-sectional study. *BMC Pulm Med* 2019;19:182.

74 Gashaw F, Bekele S, Mekonnen Y, et al. High helminthic co-infection in tuberculosis patients with undernutritional status in northeastern Ethiopia. *Infect Dis Poverty* 2019;8:88.

75 Hoyt KJ, Sarkar S, White L, et al. Effect of malnutrition on radiographic findings and mycobacterial burden in pulmonary tuberculosis. *PloS one* 2019;14:e0214011.

76 Hussien B., Hussen M.M., Seid A., et al. Nutritional deficiency and associated factors among new pulmonary tuberculosis patients of Bale Zone Hospitals, southeast Ethiopia. *BMC Res Notes* 2019;12:751.

77 Lee N, White LV, Marin FP, et al. Mid-upper arm circumference predicts death in adult patients admitted to a TB ward in the Philippines: A prospective cohort study. *PloS one* 2019;14:e0218193.

78 Mailu EW, Owiti P, Ade S, et al. Tuberculosis control activities in the private and public health sectors of Kenya from 2013 to 2017: how do they compare?. *Trans R Soc Trop Med Hyg* 2019;113:740-8.

79 Rashak HA, Sánchez-PÃ©rez HJ, Abdelbary BE, et al. Diabetes, undernutrition, migration and indigenous communities: tuberculosis in Chiapas, Mexico. *Epidemiol Infect* 2019;147:e71.

80 Ren Z, Zhao F, Chen H, et al. Nutritional intakes and associated factors among tuberculosis patients: a cross-sectional study in China. *BMC Infect Dis* 2019;19:907.

81 Wardani DWSR, Wahono EP. Predominant Determinants of Delayed Tuberculosis Sputum Conversion in Indonesia. *Indian J Commun Med* 2019;44:53-7.

82 Wessels J, Walsh CM, Nel M. Smoking habits and alcohol use of patients with tuberculosis at Standerton Tuberculosis Specialised Hospital, Mpumalanga, South Africa. *Health SA Gesondheid* 2019;24:1146.

83 White LV, Edwards T, Lee N, et al. Co-morbidities in filipino persons with tuberculosis: A cross-sectional study in urban and rural public TBDOTS facilities. *Trans R Soc Trop Med Hyg* 2019;113:S210.

84 Ma'rufi I, Ali K, Jati SK, et al. Improvement of Nutritional Status among Tuberculosis Patients by Channa striata Supplementation: A True Experimental Study in Indonesia. *BioMed Research International* 2020:1-10.

85 Musuenge BB, Poda GG, Chen PC. Nutritional Status of Patients with Tuberculosis and Associated Factors in the Health Centre Region of Burkina Faso. *Nutrients* 2020;12:10.3390/nu12092540 doi:E2540 [pii].

86 Seid G, Ayele M. Undernutrition and Mortality among Adult Tuberculosis Patients in Addis Ababa, Ethiopia. *Adv Prev Med* 2020;2020:5238010 doi:10.1155/2020/5238010 [doi].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

87 Edwards T, White LV, Lee N, et al. Effects of comorbidities on quality of life in Filipino people with tuberculosis. *The international journal of tuberculosis and lung disease : the official journal of the International Union against Tuberculosis and Lung Disease* 2020;24:712-9.

88 White LV, Edwards T, Lee N, et al. Patterns and predictors of co-morbidities in Tuberculosis: A cross-sectional study in the Philippines. *Sci Rep* 2020;10:4100.

89 Mollah A, Shrivastava P, Das DK, et al. Nutritional status of adult Tuberculosis patients in Burdwan municipality area of West Bengal. *Indian Journal of Community Health* 2020;32:438-43.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

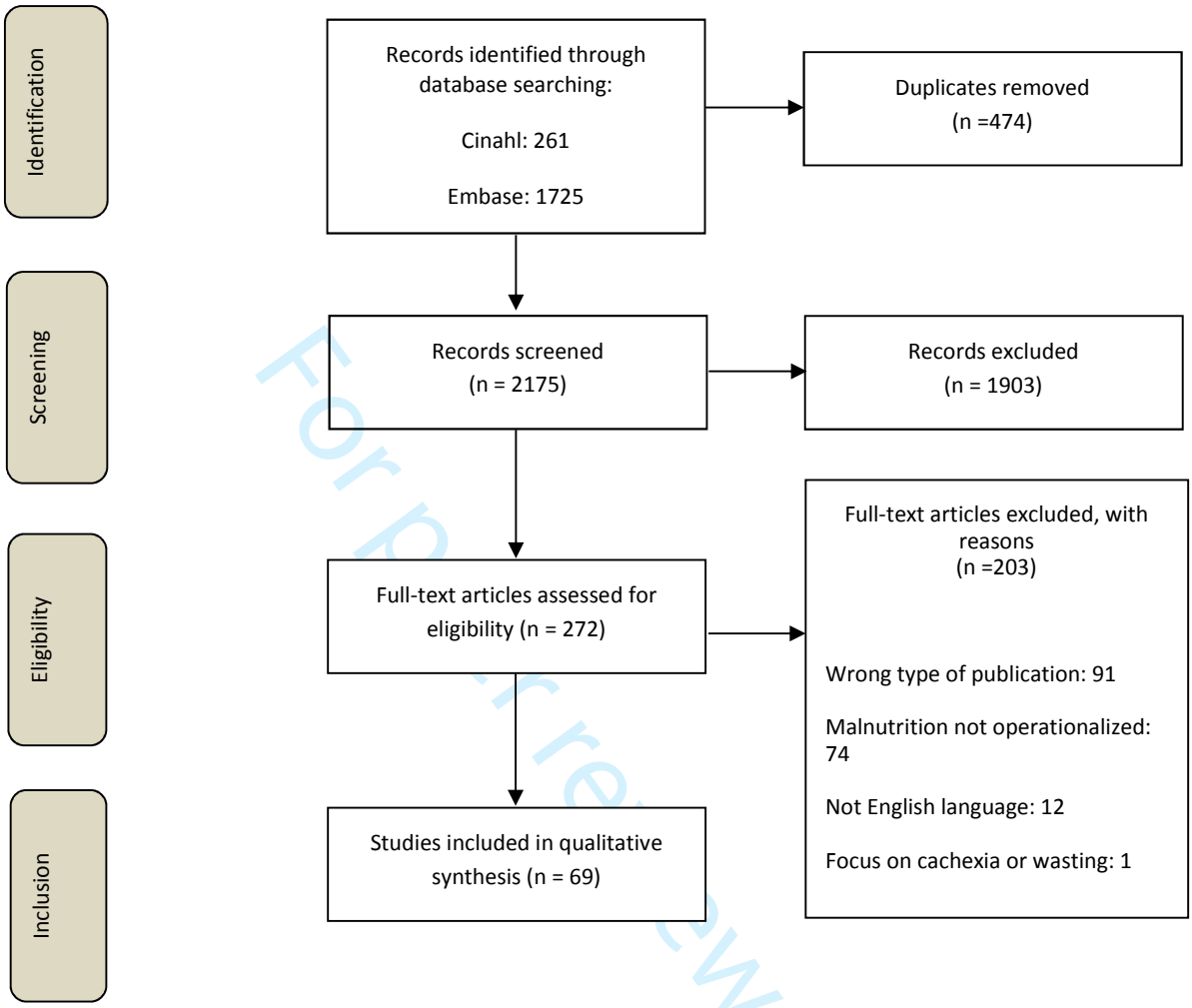


Figure 1. Flow diagram of the selection process of studies describing assessment in the context of malnutrition/undernutrition

Supplement material 1 - Details of search strategy

Embase

('malnutrition'/exp OR 'nutritional status'/exp OR Malnutrition:ab,ti OR Malnourish*:ab,ti OR 'Nutritional Deficienc*':ab,ti OR Undernutrition:ab,ti OR Undernourish*:ab,ti OR 'Nutritional Status':ab,ti OR 'Nutrition Status':ab,ti)

AND

('tuberculosis'/exp OR Tuberculosis:ab,ti OR TB:ab,ti OR tuberculoses:ab,ti) NOT ('animal'/exp NOT 'human'/exp) NOT 'child'/exp NOT 'review'/exp

Pubmed

("Malnutrition"[Mesh] OR Malnutrition [tiab] OR Malnourish* [tiab] OR Nutritional Deficienc* [tiab] OR Undernutrition [tiab] OR Undernourish* [tiab] OR "Nutritional Status"[Mesh] OR Nutritional Status [tiab] OR Nutrition Status [tiab])

AND

("Tuberculosis"[Mesh] OR Tuberculosis [tiab] OR TB [tiab] OR tuberculoses [tiab]) NOT ("animals"[MeSH] NOT "humans"[MeSH]) NOT "Child"[Mesh] NOT "Review" [Publication Type]

Cinahl

((MH "Malnutrition" OR MH "Nutritional Status" OR TI (Malnutrition OR Malnourish* OR Nutritional Deficienc* OR Undernutrition OR Undernourish* OR Nutritional Status OR Nutrition Status) OR AB (Malnutrition OR Malnourish* OR Nutritional Deficienc* OR Undernutrition OR Undernourish* OR Nutritional Status OR Nutrition Status)) AND ((MH "Tuberculosis+" OR TI (tuberculosis OR tb OR tuberculoses) OR AB (tuberculosis OR tb OR tuberculoses))

For peer review only

BMJ Open

Malnutrition assessment methods in adult patients with tuberculosis: a systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-049777.R1
Article Type:	Original research
Date Submitted by the Author:	04-Aug-2021
Complete List of Authors:	ter Beek, Lies ; University Medical Centre Groningen, Pulmonary diseases and Tuberculosis Bolhuis, Mathieu; University Medical Centre Groningen, Clinical Pharmacy and Pharmacology Jager-Wittenaar, Harriët ; Hanzehogeschool Groningen, Brijan, Rene; UMCG, Pulmonary diseases and Tuberculosis Sturkenboom, Marieke; University Medical Centre Groningen, Clincal Pharmacy and Pharmacology Kerstjens, Huib; University Medical Centre Groningen, Pulmonology de Lange, Wiel; UMCG, Pulmonary diseases and Tuberculosis; UMCG, TB center Beatrixoord Tiberi, Simon; Barts Health NHS Trust, Department of Infection; Queen Mary University of London, Blizard Institute van der Werf, Tjip S.; University Medical Centre Groningen, Department of Pulmonary Diseases & Tuberculosis Alffenaar, Jan-Willem; University of Syndney, Schoold of Pharmacy, Faculty of Medicine and Health; Westmead Hospital Akkerman, Onno W.; UMCG, Pulmonary Diseases and Tuberculosis; UMCG, TB center Beatrixoord
Primary Subject Heading:	Infectious diseases
Secondary Subject Heading:	Nutrition and metabolism
Keywords:	NUTRITION & DIETETICS, Nutrition < TROPICAL MEDICINE, Tuberculosis < INFECTIOUS DISEASES

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Malnutrition assessment methods in adult patients with tuberculosis: a systematic review

Lies ter Beek*[1,2,3], Mathieu S. Bolhuis*[4], Harriët Jager-Wittenaar[5,6], René X.D. Brijan[3], Marieke G.G. Sturkenboom[4], Huib A.M. Kerstjens[1], Wiel C.M. de Lange[1,2], Simon Tiberi[7,8], Tjip S. van der Werf[1,9], Jan-Willem C. Alffenaar[10,11,12], Onno W. Akkerman[1,2*]

1. University of Groningen, University Medical Center Groningen, Department of Pulmonary Diseases and Tuberculosis, Groningen, The Netherlands
2. University of Groningen, University Medical Center Groningen, Tuberculosis Center Beatrixoord, Haren, The Netherlands
3. Department of Health Sciences, Faculty of Science, Vrije Universiteit Amsterdam, and Amsterdam Public Health Research Institute, Amsterdam, the Netherlands
4. University of Groningen, University Medical Center Groningen, Department of Clinical Pharmacy and Pharmacology, Groningen, The Netherlands.
5. Hanze University of Applied Sciences, Research Group Healthy Ageing, Allied Health Care and Nursing, Groningen, The Netherlands
6. University of Groningen, University Medical Center Groningen, Department of Oral and Maxillofacial Surgery, Groningen, The Netherlands
7. Department of Infection, Barts Health NHS Trust, London, United Kingdom
8. Blizard Institute, Queen Mary University of London, London, United Kingdom
9. University of Groningen, University Medical Center Groningen, Department of Internal Medicine, Groningen, The Netherlands
10. University of Sydney, Faculty of Medicine and Health, School of Pharmacy, Sydney, Australia
11. Westmead Hospital, Sydney, Australia
12. Marie Bashir Institute for Infectious Diseases and Biosecurity, University of Sydney, Sydney, Australia

***both authors contributed equally to this manuscript/study**

Running head: review of malnutrition assessment in TB

Word count summary: 114

For peer review only

Word count text: 2536

Number of references: 95

Number of tables: 4

Number of figures: 1

Key words for indexing: tuberculosis, nutritional assessment, malnutrition, undernutrition

***corresponding author**

Onno W. Akkerman, MD, PhD

Tuberculosis Center Beatrixoord

University Medical Center Groningen, PO Box 30002, 9750 RA Haren, The Netherlands

e-mail: o.w.akkerman@umcg.nl

phone: +31 50 3610857

fax: +31 50 3618152

For peer review only

ABSTRACT

Objectives

Malnutrition is associated with a twofold higher risk of dying in patients with tuberculosis (TB) and considered an important potentially reversible risk factor for failure of TB treatment. The construct of malnutrition has three domains: intake or uptake of nutrition; body composition; and physical and cognitive function. The objectives of this systematic review are to identify malnutrition assessment methods, and to quantify how malnutrition assessment methods capture the international consensus definition for malnutrition, in patients with TB.

Design

Different assessment methods were identified. We determined the extent of capturing of the three domains of malnutrition, i.e., intake or uptake of nutrition, body composition and physical and cognitive function. **Results**

Seventeen malnutrition assessment methods were identified in 69 included studies. In 53/69 (77%) of studies, BMI was used as the only malnutrition assessment method. Three out of 69 studies (4%) used a method that captured all three domains of malnutrition.

Conclusions

Our study focused on published articles. Implementation of new criteria takes time, which may take longer than the period covered by this review. Most TB patients are assessed for only one aspect of the conceptual definition of malnutrition. The use of international consensus criteria is recommended to establish uniform diagnostics and treatment of malnutrition.

Trial registration

PROSPERO, accessible via https://www.crd.york.ac.uk/prospero/ with number CRD42019122832

ARTICLE SUMMARY

Strengths and limitations of this study

- PUBMED, CINAHL, and EMBASE were systematically searched for studies in the last decade, to prevent older studies describing outdated supportive care strategies to be eligible for inclusion.
- As there is no instrument for 'risk of bias assessment' of studies on diagnosis (e.g. malnutrition assessment), we evaluated risk of bias by scoring the presence of essential components required for adequate assessment and reporting of malnutrition.
- To report on how malnutrition was assessed in studies on TB, we quantified how malnutrition assessment methods capture the international consensus definition for malnutrition, in patients with TB.

For peer review only

- Implementation of new criteria in study protocols takes time. This implementation might take longer than the time period used in this review.

For peer review only

Background

In 2019, tuberculosis (TB) was the worldwide leading cause of death from a single infectious agent, with 10 million new patients, and 1.2 million deaths [1]. TB prevalence is up to 20 times higher among people living in low-income countries compared to high-income countries [2]. TB, poverty and malnutrition are closely linked. Prevalence of malnutrition in patients with TB is studied to be 50 to 57%, and malnutrition is associated with a twofold risk of dying [3-9].

As hunger-related malnutrition caused by food-insecurity impacts the immune system [1, 10, 11], malnutrition is an important risk factor for re-activation of TB, with a reported 27% attributable risk [1, 12]. Malnutrition is considered an important, potentially reversible risk factor for treatment failure [13]. Therefore, a better understanding in assessing malnutrition in TB patients is urgently needed [14].

The European Society for Clinical Nutrition and Metabolism (ESPEN) defines malnutrition as 'a state resulting from lack of intake or uptake of nutrition that leads to altered body composition (decreased fat-free mass) and body cell mass leading to diminished physical and mental function and impaired clinical outcome from disease' [10]. In patients with TB, two types of malnutrition may co-exist: malnutrition without disease [10]; and disease-related malnutrition once active disease has developed. The latter is often driven by a combination of loss of appetite, malabsorption, and/or inflammation-driven catabolism [10, 11]. A low body mass index (BMI) is a characteristic for chronic malnutrition [15]. However, since malnutrition leads to loss of fat-free mass in all individuals, including those who are overweight or obese, patients with either a normal or high BMI may be malnourished as well [10]. In 2015, ESPEN published their first consensus on diagnostic criteria for malnutrition [15], followed by the Global Leadership Initiative on Malnutrition (GLIM) criteria in 2018, which were established by ESPEN, the American Society for Parenteral and Enteral Nutrition (ASPEN), la Federación Latino Americano de Terapia Nutriconal, Nutricion Clinica y Metabolismo (FELANPE), and the Parenteral and Enteral Nutrition Society of Asia (PENSA) [16].

It was not until 2013 that the World Health Organization (WHO) presented their first guideline on nutritional care and support for TB patients. In this guideline, the WHO stressed that all patients with active TB receive individualized nutritional assessment and management, including dietary counselling and nutritional interventions, to improve nutritional status and consequently, prevent TB treatment failure [17]. Nutritional assessment is a necessary step in the 'nutritional care process', enabling the professional to design a treatment plan together with the patient [18]. However, the 2013 WHO guideline only refers to BMI as a method of assessing malnutrition for adults. The important issue of lack of consistency and understanding of how malnutrition can be assessed in patients with

For peer review only

TB results from, the complexity of TB being on the one hand strongly associated with malnutrition and on the other by the vertical and siloed nature of TB programmes.

The primary aim of this review is to identify malnutrition assessment methods that are used in adult TB patients. The secondary aim is to quantify how malnutrition assessment methods capture the international consensus definition for malnutrition in this population.

Methods

The protocol for this review was registered at PROSPERO with number CRD42019122832. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement was used [19].

Search strategy

In September 2020, PUBMED, CINAHL, and EMBASE were systematically searched for studies in any language. The search term consisted of a domain describing 'malnutrition' and a domain describing 'tuberculosis', whilst excluding *in vitro*, animal, and paediatric studies. Details of the search strategy can be found in the Supplement material 1.

Inclusion and exclusion criteria

Studies in the English language that aimed to assess malnutrition and described a method for assessment of malnutrition in patients with microbiologically confirmed or clinically diagnosedTB, published between January 1st, 2009 until September 18th, 2020 were considered eligible for inclusion. Since the aim of the review is to identify how malnutrition is assessed in current research, 2009 was chosen as a starting point. Restricting the search to the last decade will prevent older studies describing outdated supportive care strategies to be eligible for inclusion.

Only studies that focused on adult TB patients were included, because the assessment of malnutrition in children requires different methods than in adults as children are growing, and the criteria for measuring their nutritional status differ per by age [20].

Reviews and study protocols were excluded since they do not present original data. Case reports and abstracts/posters were also excluded since the information provided on methods used in this type of publication is considered too limited.

Study selection

Screening of title and abstract was done independently by two authors using the Rayyan web application [21]. Evaluation of the screening of the first ten articles was used to define the criteria that

For peer review only

determine which studies were eligible for final inclusion. Final inclusion was based on an independent judgement of the full text of both authors. Disagreements about inclusion were resolved through discussion and if consensus was not reached, a third author was consulted, which resulted in consensus in all cases.

Patient and public involvement

No patient involved

Data collection

Data collection was performed by two authors independently. The following characteristics were extracted from each study: citation, first author, country, years of publication and data collection, aim of the malnutrition assessment, number of included patients, number of HIV co-infected patients, disease location, drug susceptibility of the *M. tuberculosis* isolate.

Data analysis

To report on how malnutrition was assessed in studies on TB, we determined the extent of capturing of the three domains of malnutrition, malnutrition included in the ESPEN conceptual definition of malnutrition, i.e., intake or uptake of nutrition (Domain A), body composition (Domain B), and physical and cognitive function (Domain C)[10, 22, 23].

Domain A was considered to be covered to some extent (+) if the method addressed nutritional intake or uptake at all. It was considered to be covered extensively (++) if the method addressed nutritional intake or uptake in depth. For domain B, weight change, BMI, and anthropometric measurements, such as skinfold measurements were considered covering domain B to some extent (+). It was considered extensively addressed (++) if the method included identification of muscle mass, lean mass, or fat-free mass. Domain C was considered to be covered to some extent (+) if functionality was addressed. Domain C was considered extensively covered (++) if physical (e.g. handgrip strength), mental and cognitive function tests were performed, or questions about activities of daily living were addressed.

As micronutrient or trace elements in serum are not representative for intake or uptake of protein or energy [10], laboratory tests were not considered to attribute to any of the malnutrition domains. Serum albumin and CRP are parameters for inflammation but are not related to parameters of protein/energy intake or functionality, nor do they represent body composition and were therefore not taken into consideration. In addition, while prealbumin (i.e. transthyretin) is sensitive for changes in protein and energy intake, this marker is influenced by inflammation activity [10].

As there is no instrument for 'risk of bias assessment' of studies on diagnosis (e.g. malnutrition

For peer review only

assessment), we evaluated risk of bias by scoring the presence of essential components required for adequate assessment and reporting of malnutrition. The risk of bias in included studies was assessed by rating the following four characteristics:

- rationale for the assessment of malnutrition

-malnutrition was assessed and clear cut-off points were described in the method section. For example: 'malnutrition was assessed as BMI < 18.5 kg/m^2

-malnutrition was reported in the results section.

-the results with regard to malnutrition were reflected upon in the discussion section.

The characteristics were graded for quality of the study. Risk of bias was rated by - (meaning high risk of bias), + (medium risk of bias), or ++ (low risk of bias). The scores were added and the total was translated into very low (≥7 plusses), low (5-6 plusses), medium (3-4 plusses) or high risk of bias (≤2 plusses).

Results

The search resulted in 2175 studies after removal of duplicates. After screening by title and abstract, 272 studies were selected for a full text eligibility check. Review of the full text resulted in inclusion of 69 studies[3, 24-91]. A flow diagram of the selection process is visualized in **Figure 1**.

In total, among 69 studies, 17 different methods were used to assess malnutrition. Four studies used multiple (i.e., two to four) methods to assess malnutrition. Four studies used multiple criteria but integrated these together into one method to perform an assessment. In 53/69 (77%) of the studies, BMI was used as single assessment method. Among these studies eight different cut-off points were used, 34/53 studies (64%) used BMI < 18.5 kg/m^2. In seven studies, no cut-off values for BMI were described. **Table 1** shows all the cut-off values of BMI used as single method.

Capturing of the domains of malnutrition

Table 2 presents the capturing of the domains of malnutrition per assessment method[3, 24-91]. Uptake or intake of nutrition (A) was addressed to some extent by four methods that were used in five of the 69 studies. Body composition (B) was addressed to some extent by all but two studies, 67/69 (97%). Two methods did not address domain B, as these were methods that used self-reporting of diet quality and a food frequency questionnaire as assessment methods. Physical and/or cognitive functionality (C) was addressed to some extent by only 3/69 studies (4%).

Only two of the 17 different assessment methods that were used in the 69 studies captured all three domains of the definition, i.e., the Subjective Global Assessment (SGA) was used in two studies and the Mini Nutritional Assessment (MNA) was used in one study. Forty-one studies (59%) assessed malnutrition for the purpose of their primary aim. The three studies that used a method that

For peer review only

captured all three domains of the definition, had a primary aim related to malnutrition.

A descriptive subanalysis, as shown in **Table 3**, was performed to compare studies regarding their use of assessment tools, by their period of data collection[3, 24-91]. No differences were found between five-year periods regarding the use of BMI as the only assessment method. The studies that used methods that capture all three domains of the ESPEN definition of malnutrition did not describe when their data was collected Of the 69 included studies, 18 studies (26%) were on inpatients, 26 studies (38%) on outpatients, 7 (10%) on both type of patients and in 18 studies (26%) the type of patients was not described.

Risk of bias assessment

The supplement material 2 shows the risk of bias assessment and details of the 69 included studies[3, 24-91]. Twenty-one of the 69 studies (30%) did not describe the rationale for the assessment of malnutrition. Fifteen of the 69 studies (22%) described their assessment method without clear cut-off values. Risk of bias was very low in 24 studies (35%), low in 20 studies (29%), medium in 20 studies (29%) and high in five studies (7%).

Discussion

This review showed that most TB patients are assessed for only one aspects of the construct of malnutrition. BMI is often used assessment method for malnutrition in studies with TB patients, even though it only partly covers just one malnutrition domain (domain B). In addition, while some studies did not report a cut-off value for BMI, in other studies different cut-off values for BMI were used, therefore making it difficult to compare these studies. The use of BMI could be justified from a public health perspective, since a low BMI is a characteristic of chronic malnutrition that involves loss of both fat and muscle tissue[15]. However, the use of BMI alone for assessing malnutrition in this population is debatable. In clinical settings, disease-related malnutrition is the predominant type of malnutrition. Disease-related malnutrition is a (sub)acute condition, in which loss of weight and muscle/fat-free mass does not automatically result in a low BMI, while loss of weight and fat-free mass are related to poor clinical outcomes including increased morbidity and mortality [15, 92]. With the current global overweight and obesity epidemic, patients with catabolic diseases such as TB may lose more than 20% of their weight and muscle mass within three to six months, and still show BMI values at or above normal range [93]. When assessing malnutrition in TB patients solely based on BMI, these patients will not be identified when they are malnourished, despite the therapeutic and prognostic implications of malnutrition[94].

This review indicates that the international criteria for the assessment of malnutrition have not yet found their way into studies with TB patients. The 2013 WHO guideline on nutritional care only refers

For peer review only

to BMI as a method of assessing malnutrition for adults, which may contribute to the status quo. BMI is by far the most frequently used assessment method for malnutrition (77%) in studies with TB patients. Only a few (4%) of the studies that reported on the assessment methods addressed all domains of the conceptual definition of malnutrition. For the studies performed after the ESPEN criteria of 2015 [95] and the GLIM criteria of 2018 [16], this may be explained by the fact that the WHO does not yet refer to these criteria in their communications. The GLIM criteria have been developed for global use and is therefore recommended to be used in any setting for all TB patients. The GLIM framework does not include the domain of functionality as criterion.

There are some limitations in our study. First, our study focused on published articles and not on the underlying study protocol. In some cases, detailed information on the assessment and operationalisation of malnutrition might be available in the study protocol, however it was not addressed in the article making it unavailable to the public domain. Second, implementation of new criteria in study protocols takes time. This implementation might take longer than the time period used in this review. Third: research and clinical practice are different settings and the results from our review are not a reflection of clinical practice. Nevertheless, we postulate that BMI is used the most commonly used method in clinical practice since the WHO recommends that BMI is the method to assess undernutrition/malnutrition.

Awareness of the presence of malnutrition and concept of malnutrition assessment in health care professionals working with TB patients needs improvement. Future studies regarding malnutrition assessment in TB patients should aim at implementation of international consensus criteria regarding malnutrition assessment. Using the same terminology for malnutrition may make a difference to our patients, improve outcomes as well as reduce chronic sequelae and help to end TB.

It should be stressed that we need to agree on using standardized methods for malnutrition assessment and interventions. However, malnutrition assessment should always be preceded by malnutrition screening with a validated tool (Supplement material 3). [16]

In conclusion, most studies in adult TB patients did not describe their assessment method for malnutrition. Most TB patients are assessed for only one or two aspects of the conceptual definition of malnutrition. Various methods for assessing of malnutrition have been used, and only a very small proportion of the published studies on TB used an assessment method that fully reflects the definition of malnutrition. The use of international consensus criteria is recommended to establish systematic and uniform diagnostics and treatment of malnutrition.

For peer review only

Ethics approval statement

Not applicable for this systematic review

Funding statement

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing Interests statement

H. Jager-Wittenaar was co-developer of the Patient-Generated Subjective Global Assessment©-based Pt-Global© app/web tool.

Other authors have no conflict of interest to disclose.

Author contributions

Lies ter Beek: conception and design, data collection, acquisition, analysis, interpretation of data; drafting

Mathieu S. Bolhuis: conception and design, data collection, acquisition, analysis, interpretation of data; revising draft critically for important intellectual content

Harriët Jager-Wittenaar: interpretation of data and revising draft critically for important intellectual content

René X.D. Brijan: acquisition, analysis, interpretation of data; drafting

Marieke G.G. Sturkenboom: interpretation of data, and revising draft critically for important intellectual content

Huib A.M. Kerstjens: interpretation of data, and revising draft critically for important intellectual content

Wiel C.M. de Lange: interpretation of data, and revising draft critically for important intellectual content

Simon Tiberi: interpretation of data, and revising draft critically for important intellectual content

Tjip S. van der Werf: interpretation of data, and revising draft critically for important intellectual content

Jan-Willem C. Alffenaar: interpretation of data, and revising draft critically for important intellectual content

Onno W. Akkerman: conception and design, interpretation of data; revising draft critically for important intellectual content

Data availability

For peer review only

No additional data available

Acknowledgements

We thank Martine Sealy for her contribution to the conceptualization and visualization of this review. We thank Giovanni Sotgiu for his feedback on the manuscript. We thank the Central Medical Library of the University of Groningen for their help with designing our search strategy.

Figure legend

Figure 1. Flow diagram of the selection process of studies describing assessment in the context of malnutrition

Table 1 Cut-off values BMI exclusively used

Cut-off value	Studies using only BMI for assessment of malnutrition n= 53
≤ 20.0 kg/m^2	1
< 20.0 kg/m^2	2
< 18.5 kg/m^2	34
≤ 18.5 kg/m^2	3
< 18.49 kg/m^2	1
< 18.4 kg/m^2	1
< 17.0 kg/m^2	2
< 16.0 kg/m^2	2
No clear cut-off value described in study	7

For peer review only

Table 2 Comparison of 17 different assessment methods by malnutrition domain capturing

DOMAINS Description of domain	A intake or uptake of nutrition	B body composition	C physical and cognitive function	Total
PIBW, BMI, Albumin, TLC, Cholesterol, Hb	-	+	-	+
Weight change	-	+	-	+
BMI & MUAC	-	+	-	+
Self-report of diet quality	+	-	-	+
MNA	++	+	++	+++++
MUST	+	+	-	++
AMA	-	++	-	++
BMI/Albumin	-	+	-	+
BMI, MUAC, TSF, MAMC, Hb, Ht, albumin, total protein, globulin, iron fixation capacity, retinol, tocopherol Zn, Se, Fe	-	+	-	+
MAMC	-	++	-	++
underweight	-	+	-	+
TSF/TST	-	+	-	+
SGA	++	++	++	++++++
BMI	-	+	-	+
BIA: (Body fat%)	-	+	-	+
MUAC	-	+	-	+
FFQ	++	-	-	++

Abbreviations: *PIBW: Percent Ideal Body Weight; BMI: Body Mass index; TLC: Total Lymphocyte Count; Hb: Haemoglobin; Ht: Haematocrit; MUAC: Mid Upper Arm Circumference; MNA: Mini Nutritional Assessment; AMA: Arm Muscle Area; MAMC: Mid Arm Muscle Circumference; Zn: Zinc; Se: Selenium; Fe: Iron; TSF: Triceps Skin Fold ; TST: Triceps Skinfold Thickness; SGA: Subjective Global Assessment; BIA: Bio-electrical Impedance Analysis; FFQ: Food Frequency Questionnaire*

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3 Subanalysis of malnutrition assessment methods

Year of start data collection	Number of studies (n=69)	BMI as the only assessment method		Use of an assessment method that attributes to 3 domains	
		N	%		%
2000-2004	9	9	89	-	
2005-2009	12	7	58	-	
2010-2014	14	13	93	-	
2015-2019	19	15	79	-	
No available data on year of data collection	15	10	67	3	20
Total	69	53	77	3	4

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1 World Health Organisation (WHO). Global Tuberculosis Report 2020 2020;Geneva, WHO.

2 World Health Organization (WHO). Addressing poverty in TB control - options for national TB control programmes 2005;Geneva, WHO.

3 Bhargava A, Chatterjee M, Jain Y, et al. Nutritional status of adult patients with pulmonary tuberculosis in rural central India and its association with mortality. *PLoS One* 2013;8:e77979 doi:10.1371/journal.pone.0077979 [doi].

4 Nthiga I, David-Kigaru D, Mugendi J, et al. The nutritional status of pulmonary tuberculosis patients aged 25-44 years attending tuberculosis clinic at Lodwar County and Referral Hospital, Turkana County, Kenya 2017;2:2455-4898.

5 Wondmieneh A, Gedefaw G, Getie A, et al. Prevalence of undernutrition among adult tuberculosis patients in Ethiopia: A systematic review and meta-analysis. *J Clin Tuberc Other Mycobact Dis* 2020;22:100211 doi:10.1016/j.jctube.2020.100211 [doi].

6 Zachariah R, Spielmann MP, Harries AD, et al. Moderate to severe malnutrition in patients with tuberculosis is a risk factor associated with early death. *Trans R Soc Trop Med Hyg* 2002;96:291-4.

7 Dodor E. Evaluation of nutritional status of new tuberculosis patients at the effia-nkwanta regional hospital. *Ghana Med J* 2008;42:22-8.

8 SanchitaSubedi, Mehta R, PushpaParajuli, et al. Nutritional Status of Patients with Pulmonary Tuberculosis receiving Anti-Tuberculosis Treatment at BP Koirala Institute of Health Sciences, Nepal 2019.

9 Bhargava A, Benedetti A, Oxlade O, et al. Undernutrition and the incidence of tuberculosis in India: national and subnational estimates of the population-attributable fraction related to undernutrition. *Natl Med J India* 2014;27:128-33.

10 Cederholm T, Barazzoni R, Austin P, et al. ESPEN guidelines on definitions and terminology of clinical nutrition. *Clin Nutr* 2017;36:49-64 doi:S0261-5614(16)31242-0 [pii].

11 Sobotka L, Allison SP, Forbes A, et al. Basics in clinical nutrition. Prague, Czech Republic: Publishing House Galen 2011.

12 Dheda K, Barry CE,3rd, Maartens G. Tuberculosis. *Lancet* 2016;387:1211-26 doi:10.1016/S0140-6736(15)00151-8 [doi].

13 Choi R, Jeong BH, Koh WJ, et al. Recommendations for Optimizing Tuberculosis Treatment: Therapeutic Drug Monitoring, Pharmacogenetics, and Nutritional Status Considerations. *Ann Lab Med* 2017;37:97-107 doi:10.3343/alm.2017.37.2.97 [doi].

14 World Health Organization (WHO). The End TB Strategy 2014;Geneva, WHO.

15 Cederholm T, Bosaeus I, Barazzoni R, et al. Diagnostic criteria for malnutrition - An ESPEN Consensus Statement. *Clin Nutr* 2015;34:335-40 doi:S0261-5614(15)00075-8 [pii].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16 Cederholm T, Jensen GL, Correia MITD, et al. GLIM criteria for the diagnosis of malnutrition - A consensus report from the global clinical nutrition community. *J Cachexia Sarcopenia Muscle* 2019;10:207-17 doi:10.1002/jcsm.12383 [doi].

17 World Health Organisation (WHO). Nutritional care and support for patients with tuberculosis 2013;Geneva, WHO doi:NBK189867 [bookaccession].

18 Lacey K, Pritchett E. Nutrition Care Process and Model: ADA adopts road map to quality care and outcomes management. *J Am Diet Assoc* 2003;103:1061-72 doi:10.1053/jada.2003.50564 [doi].

19 Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;6:e1000097 doi:10.1371/journal.pmed.1000097 [doi].

20 Turck D, Braegger CP, Colombo C, et al. ESPEN-ESPGHAN-ECFS guidelines on nutrition care for infants, children, and adults with cystic fibrosis. *Clin Nutr* 2016;35:557-77 doi:10.1016/j.clnu.2016.03.004 [doi].

21 Ouzzani M, Hammady H, Fedorowicz Z, et al. Rayyan-a web and mobile app for systematic reviews. *Syst Rev* 2016;5:210,016-0384-4 doi:10.1186/s13643-016-0384-4 [doi].

22 Ter Beek L, Vanhauwaert E, Slinde F, et al. Unsatisfactory knowledge and use of terminology regarding malnutrition, starvation, cachexia and sarcopenia among dietitians. *Clin Nutr* 2016;35:1450-6 doi:S0261-5614(16)30012-7 [pii].

23 Sealy MJ, Nijholt W, Stuiver MM, et al. Content validity across methods of malnutrition assessment in patients with cancer is limited. *J Clin Epidemiol* 2016;76:125-36 doi:10.1016/j.jclinepi.2016.02.020 [doi].

24 Mollah A, Shrivastava P, Das DK, et al. Nutritional status of adult Tuberculosis patients in Burdwan municipality area of West Bengal. *Indian Journal of Community Health* 2020;32:438-43.

25 White LV, Edwards T, Lee N, et al. Patterns and predictors of co-morbidities in Tuberculosis: A cross-sectional study in the Philippines. *Sci Rep* 2020;10:4100.

26 Edwards T, White LV, Lee N, et al. Effects of comorbidities on quality of life in Filipino people with tuberculosis. *The international journal of tuberculosis and lung disease : the official journal of the International Union against Tuberculosis and Lung Disease* 2020;24:712-9.

27 Seid G, Ayele M. Undernutrition and Mortality among Adult Tuberculosis Patients in Addis Ababa, Ethiopia. *Adv Prev Med* 2020;2020:5238010 doi:10.1155/2020/5238010 [doi].

28 Musuenge BB, Poda GG, Chen PC. Nutritional Status of Patients with Tuberculosis and Associated Factors in the Health Centre Region of Burkina Faso. *Nutrients* 2020;12:10.3390/nu12092540 doi:E2540 [pii].

29 Ma'rufi I, Ali K, Jati SK, et al. Improvement of Nutritional Status among Tuberculosis Patients by Channa striata Supplementation: A True Experimental Study in Indonesia. *BioMed Research International* 2020:1-10.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

30 White LV, Edwards T, Lee N, et al. Co-morbidities in filipino persons with tuberculosis: A cross-sectional study in urban and rural public TBDOTS facilities. *Trans R Soc Trop Med Hyg* 2019;113:S210.

31 Wessels J, Walsh CM, Nel M. Smoking habits and alcohol use of patients with tuberculosis at Standerton Tuberculosis Specialised Hospital, Mpumalanga, South Africa. *Health SA Gesondheid* 2019;24:1146.

32 Wardani DWSR, Wahono EP. Predominant Determinants of Delayed Tuberculosis Sputum Conversion in Indonesia. *Indian J Commun Med* 2019;44:53-7.

33 Ren Z, Zhao F, Chen H, et al. Nutritional intakes and associated factors among tuberculosis patients: a cross-sectional study in China. *BMC Infect Dis* 2019;19:907.

34 Rashak HA, Sánchez-PÃ©rez HJ, Abdelbary BE, et al. Diabetes, undernutrition, migration and indigenous communities: tuberculosis in Chiapas, Mexico. *Epidemiol Infect* 2019;147:e71.

35 Mailu EW, Owiti P, Ade S, et al. Tuberculosis control activities in the private and public health sectors of Kenya from 2013 to 2017: how do they compare?. *Trans R Soc Trop Med Hyg* 2019;113:740-8.

36 Lee N, White LV, Marin FP, et al. Mid-upper arm circumference predicts death in adult patients admitted to a TB ward in the Philippines: A prospective cohort study. *PloS one* 2019;14:e0218193.

37 Hussien B., Hussen M.M., Seid A., et al. Nutritional deficiency and associated factors among new pulmonary tuberculosis patients of Bale Zone Hospitals, southeast Ethiopia. *BMC Res Notes* 2019;12:751.

38 Hoyt KJ, Sarkar S, White L, et al. Effect of malnutrition on radiographic findings and mycobacterial burden in pulmonary tuberculosis. *PloS one* 2019;14:e0214011.

39 Gashaw F, Bekele S, Mekonnen Y, et al. High helminthic co-infection in tuberculosis patients with undernutritional status in northeastern Ethiopia. *Infect Dis Poverty* 2019;8:88.

40 Feleke BE, Feleke TE, Biadglegne F. Nutritional status of tuberculosis patients, a comparative cross-sectional study. *BMC Pulm Med* 2019;19:182.

41 da Silva LF, Skupien EC, Lazzari T, et al. Advanced glycation end products (AGE) and receptor for AGE (RAGE) in patients with active tuberculosis, and their relationship between food intake and nutritional status. *PloS one* 2019;14:e0213991.

42 Chebrolu P., Laux T., Chowdhury S., et al. The risk of refeeding syndrome among severely malnourished tuberculosis patients in Chhattisgarh, India. *Indian J Tuberc* 2019:https://doi.org/10.1016/j.ijtb.2019.03.004.

43 Benzekri NA, Sambou JF, Tamba IT, et al. Nutrition support for HIV-TB co-infected adults in Senegal, West Africa: A randomized pilot implementation study. *PloS one* 2019;14:e0219118.

44 Abdullahi OA, Ngari MM, Sanga D, et al. Mortality during treatment for tuberculosis; a review of surveillance data in a rural county in Kenya. *PloS one* 2019;14:e0219191.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

45 Skupien EC, Lazzari T, Coutinho SE, et al. The relation between leptin and inflammatory markers with respiratory and peripheral muscle strength in tuberculosis: A case-control study. *Clin Respir J* 2018;12:2559-65.

46 Cheng W, Zhang S, Li Y, et al. Intestinal tuberculosis: clinico-pathological profile and the importance of a high degree of suspicion. *Trop Med Int Health* 2019;24:81-90.

47 Lazzari T.K., Forte G.C., Silva D.R. Nutrition Status Among HIV-Positive and HIV-Negative Inpatients with Pulmonary Tuberculosis. *Nutr.Clin.Prac.* 2018;33:858-64.

48 Sattler FR, Chelliah D, Wu X, et al. Biomarkers Associated with Death After Initiating Treatment for Tuberculosis and HIV in Patients with Very Low CD4 Cells. *Pathogens & immunity* 2018;3:46-62.

49 Rao VG, Bhat J, Yadav R, et al. A comparative study of the socio-economic risk factors for pulmonary tuberculosis in the Saharia tribe of Madhya Pradesh, India. *Trans R Soc Trop Med Hyg* 2018;112:272-8.

50 Pande T, Huddart S, Xavier W, et al. Prevalence of diabetes mellitus amongst hospitalized tuberculosis patients at an Indian tertiary care center: A descriptive analysis. *PloS one* 2018;13:e0200838.

51 Gurung LM, Bhatt LD, Karmacharya I, et al. Dietary Practice and Nutritional Status of Tuberculosis Patients in Pokhara: A Cross Sectional Study. *Frontiers in nutrition* 2018;5:1-6.

52 Piparva KG, Jansari G, Singh AP. Evaluation of treatment outcome and adverse drug reaction of directly observed treatment (DOT) plus regimen in multidrug-resistant tuberculosis (MDR-TB) patients at district tuberculosis centre Rajkot. *Perspectives in clinical research* 2018;9:165-9.

53 Hochberg NS, Sarkar S, Horsburgh CR, et al. Comorbidities in pulmonary tuberculosis cases in Puducherry and Tamil Nadu, India: Opportunities for intervention. *PloS one* 2017;12:e0183195.

54 Bhat J, Rao VG, Sharma RK, et al. Investigation of the risk factors for pulmonary tuberculosis: A case-control study among Saharia tribe in Gwalior district, Madhya Pradesh, India. *Indian J Med Res* 2017;146:97-104.

55 Abdelbary BE, Garcia-Viveros M, Ramirez-Oropesa H, et al. Predicting treatment failure, death and drug resistance using a computed risk score among newly diagnosed TB patients in Tamaulipas, Mexico. *Epidemiol Infect* 2017;145:3020-34.

56 Pandit A, Pandey Ak. Liver dysfunction in pulmonary tuberculosis patients on dots: A study and review. *J Gastroenterol Hepatol Res* 2016;5:2254-60.

57 Buntoro IF, Kristin E, Sumardi. Decrease of liver function after treatment of antituberculosis drugs in tuberculosis patients with malnutrition and alcohol consumption. *Int J Pharm Pharm Sci* 2016;8:269-73.

58 Araújo-Mariz C, Lopes EP, Acioli-Santos B, et al. Hepatotoxicity during Treatment for Tuberculosis in People Living with HIV/AIDS. *PloS one* 2016;11:e0157725.

59 Gebrecherkos T, Gelaw B, Tessema B. Smear positive pulmonary tuberculosis and HIV co-infection in prison settings of North Gondar Zone, Northwest Ethiopia. *BMC Public Health* 2016;16:1091.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

60 McLachlan I, Visser WI, Jordaan HF. Skin conditions in a South African tuberculosis hospital: Prevalence, description, and possible associations. *Int J Dermatol* 2016;55:1234-41.

61 Ezeamama AE, Mupere E, Oloya J, et al. Age, sex, and nutritional status modify the CD4+ T-cell recovery rate in HIV-tuberculosis co-infected patients on combination antiretroviral therapy. *Int J Infect Dis* 2015;35:73-9 doi:10.1016/j.ijid.2015.04.008 [doi].

62 Medellin-Garibay SE, Cortez-Espinosa N, Milan-Segovia RC, et al. Clinical Pharmacokinetics of Rifampin in Patients with Tuberculosis and Type 2 Diabetes Mellitus: Association with Biochemical and Immunological Parameters. *Antimicrob Agents Chemother* 2015;59:7707-14 doi:10.1128/AAC.01067-15 [doi].

63 Te Brake LH, Ruslami R, Later-Nijland H, et al. Exposure to total and protein-unbound rifampicin is not affected by malnutrition in Indonesian tuberculosis patients. *Antimicrob Agents Chemother* 2015;74:986-90 doi:AAC.03485-14 [pii].

64 Golemba AS, Ferreyra FG, Martearena RE, et al. Drug-induced hepatotoxicity and tuberculosis in a hospital from the Argentinian northeast: cross-sectional study. *Medwave* 2015;15:e6135 doi:10.5867/medwave.2015.04.6135 [doi].

65 Bacelo AC, Ramalho A, Brasil PE, et al. Nutritional Supplementation Is a Necessary Complement to Dietary Counseling among Tuberculosis and Tuberculosis-HIV Patients. *PLoS One* 2015;10:e0134785 doi:10.1371/journal.pone.0134785 [doi].

66 Kumar NSS, Hemraj SK, Dutt RA. Phase angle measurement in pulmonary tuberculosis patients and control subjects using bio-impedance analysis. *The Indian journal of tuberculosis* 2014;61:224-31.

67 Tian PW, Wang Y, Shen YC, et al. Different risk factors of recurrent pulmonary tuberculosis between Tibetan and Han populations in Southwest China. *Eur Rev Med Pharmacol Sci* 2014;18:1482-6 doi:7406 [pii].

68 Oliveira MG, Delogo KN, Oliveira,Hedi Marinho de Melo Gomes de, et al. Anemia in hospitalized patients with pulmonary tuberculosis. *Jornal brasileiro de pneumologia : publicacao oficial da Sociedade Brasileira de Pneumologia e Tisilogia* 2014;40:403-10.

69 Maeda S, Hang NT, Lien LT, et al. Mycobacterium tuberculosis strains spreading in Hanoi, Vietnam: Beijing sublineages, genotypes, drug susceptibility patterns, and host factors. *Tuberculosis (Edinb)* 2014;94:649-56 doi:S1472-9792(14)20533-6 [pii].

70 Miyata S, Tanaka M, Ihaku D. The prognostic significance of nutritional status using malnutrition universal screening tool in patients with pulmonary tuberculosis. *Nutrition journal* 2013;12:42.

71 Ismawati. R., Bambang W, Priatna DY, et al. The effect of zinc, lysine and vitamin a supplementation to increase cellular immune response of pulmonary tuberculosis patients. *J Trace Elem Med Biol* 2013;27:21.

72 Bakari M, Wamsele J, MacKenzie T, et al. Nutritional status of HIV-infected women with tuberculosis in Dar es Salaam, Tanzania. *Public health action* 2013;3:224-9.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

73 Miyata S, Tanaka M, Ihaku D. Full mini nutritional assessment and prognosis in elderly patients with pulmonary tuberculosis. *J Am Coll Nutr* 2013;32:307-11 doi:10.1080/07315724.2013.826114 [doi].

74 Chittoor G, Arya R, Farook VS, et al. Epidemiologic investigation of tuberculosis in a Mexican population from Chihuahua State, Mexico: a pilot study. *Tuberculosis (Edinburgh, Scotland)* 2013;93:S71-7.

75 Islam QS, Ahmed SM, Islam MA, et al. Beyond drugs: tuberculosis patients in Bangladesh need nutritional support during convalescence. *Public health action* 2013;3:136-40.

76 Piva SG, Costa Mda C, Barreto FR, et al. Prevalence of nutritional deficiency in patients with pulmonary tuberculosis. *J Bras Pneumol* 2013;39:476-83 doi:10.1590/S1806-37132013000400012 [doi].

77 Mupere E, Malone L, Zalwango S, et al. Lean tissue mass wasting is associated with increased risk of mortality among women with pulmonary tuberculosis in urban Uganda. *Ann Epidemiol* 2012;22:466-73.

78 Miyata S, Tanaka M, Ihaku D. Subjective global assessment in patients with pulmonary tuberculosis. *Nutr Clin Pract* 2011;26:55-60 doi:10.1177/0884533610392380 [doi].

79 Kawai K, Villamor E, Mugusi FM, et al. Predictors of change in nutritional and hemoglobin status among adults treated for tuberculosis in Tanzania. *Int J Tuberc Lung Dis* 2011;15:1380-9 doi:10.5588/ijtld.10.0784 [doi].

80 de Jong BC, Adetifa I, Walther B, et al. Differences between tuberculosis cases infected with Mycobacterium africanum, West African type 2, relative to Euro-American Mycobacterium tuberculosis: an update. *FEMS Immunol Med Microbiol* 2010;58:102-5 doi:10.1111/j.1574-695X.2009.00628.x [doi].

81 Podewils LJ, Holtz T, Riekstina V, et al. Impact of malnutrition on clinical presentation, clinical course, and mortality in MDR-TB patients. *Epidemiol Infect* 2011;139:113-20 doi:10.1017/S0950268810000907 [doi].

82 Warmelink I, van Altena R, ten Hacken N, et al. Nutritional status and vitamin D3 during antimicrobial treatment. *Lancet* 2011;377:1407-8.

83 Mupere E, Zalwango S, Chiunda A, et al. Body composition among HIV-seropositive and HIV-seronegative adult patients with pulmonary tuberculosis in Uganda. *Ann Epidemiol* 2010;20:210-6.

84 Singla R, Sharma SK, Mohan A, et al. Evaluation of risk factors for antituberculosis treatment induced hepatotoxicity. *Indian J Med Res* 2010;132:81-6.

85 Gambhir HS, Kaushik RM, Kaushik R, et al. Tobacco smoking-associated risk for tuberculosis: a case-control study. *Int Health* 2010;2:216-22.

86 Pakasi TA, Karyadi E, Suratih NMD, et al. Zinc and vitamin A supplementation fails to reduce sputum conversion time in severely malnourished pulmonary tuberculosis patients in Indonesia. *Nutr J* 2010;9:41.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

87 Khoharo HK, Ansari S, Siddiqui AA, et al. Standard antituberculosis drug induced hepatotoxicity: Do the risk factors matter?. *J Liaquat Uni Med Health Sci* 2010;9:84-7.

88 Kim JS, Wilson JM, Lee SR. Dietary implications on mechanisms of sarcopenia: roles of protein, amino acids and antioxidants. *J Nutr Biochem* 2010;21:1-13 doi:10.1016/j.jnutbio.2009.06.014; 10.1016/j.jnutbio.2009.06.014.

89 Ulasli SS, Ulubay G, Arslan NG, et al. **Characteristics and Outcomes of End-Stage Renal Disease Patients With Active Tuberculosis Followed in Intensive Care Units**. *Saudi J Kidney Dis Transpl* 2009;20:254-9.

90 Pakasi TA, Karyadi E, Dolmans WM, et al. Malnutrition and socio-demographic factors associated with pulmonary tuberculosis in Timor and Rote Islands, Indonesia. *Int J Tuberc Lung Dis* 2009;13:755-9.

91 Martins N, Morris P, Kelly PM. Food incentives to improve completion of tuberculosis treatment: randomised controlled trial in Dili, Timor-Leste. *BMJ* 2009;339:b4248 doi:10.1136/bmj.b4248 [doi].

92 Correia MI, Waitzberg DL. The impact of malnutrition on morbidity, mortality, length of hospital stay and costs evaluated through a multivariate model analysis. *Clin Nutr* 2003;22:235-9 doi:S0261561402002157 [pii].

93 Gonzalez MC, Correia MITD, Heymsfield SB. A requiem for BMI in the clinical setting. *Curr Opin Clin Nutr Metab Care* 2017;20:314-21 doi:10.1097/MCO.0000000000000395 [doi].

94 Jensen GL, Mirtallo J, Compher C, et al. Adult starvation and disease-related malnutrition: a proposal for etiology-based diagnosis in the clinical practice setting from the International Consensus Guideline Committee. *JPEN J Parenter Enteral Nutr* 2010;34:156-9 doi:10.1177/0148607110361910; 10.1177/0148607110361910.

95 Cederholm T, Bosaeus I, Barazzoni R, et al. Diagnostic criteria for malnutrition - An ESPEN Consensus Statement. *Clin Nutr* 2015;34:335-40 doi:10.1016/j.clnu.2015.03.001 [doi].

First author	
3, 24-91	

For peer review only

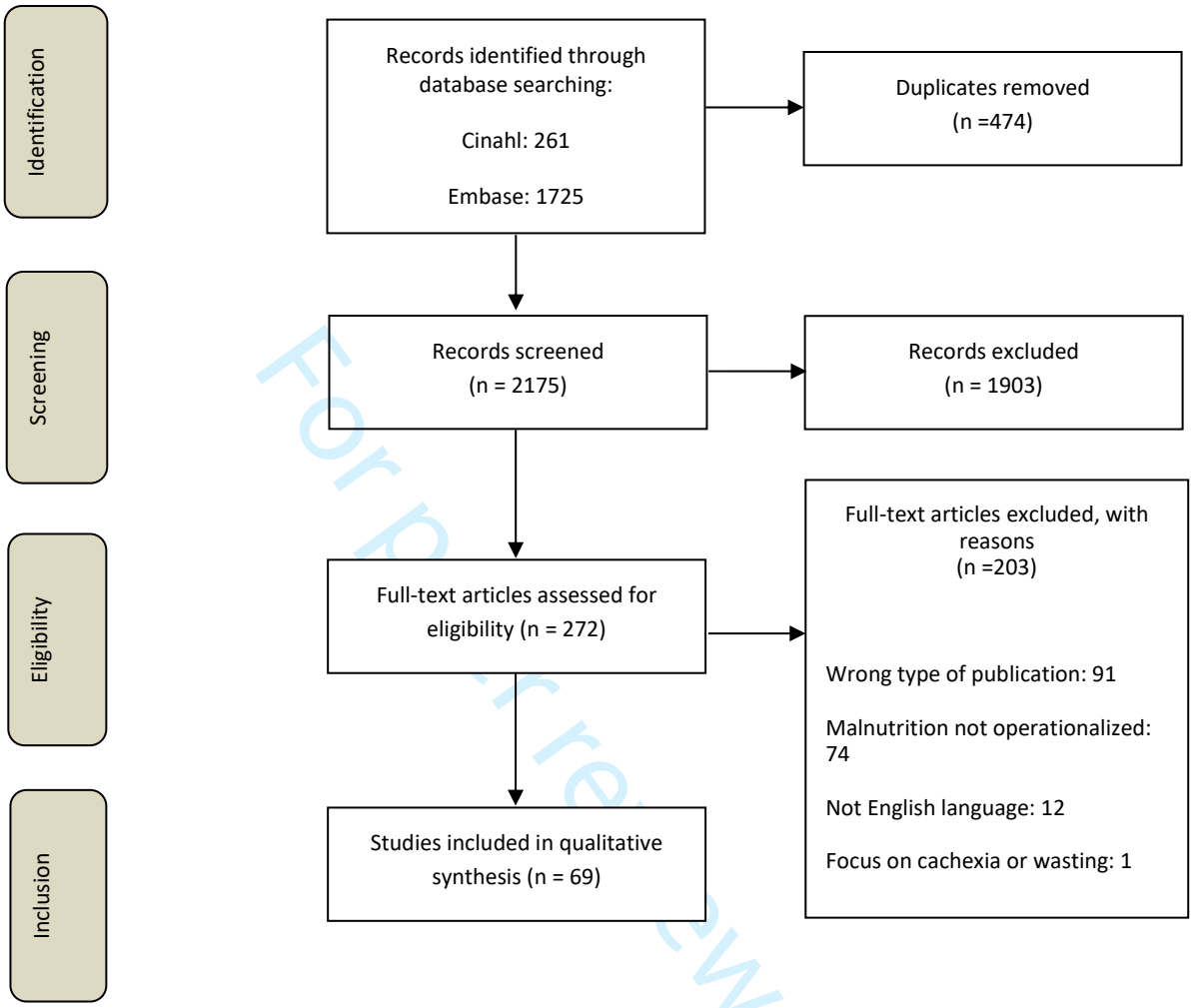


Figure 1. Flow diagram of the selection process of studies describing assessment in the context of malnutrition

Supplement material 1 - Details of search strategy

Embase

('malnutrition'/exp OR 'nutritional status'/exp OR Malnutrition:ab,ti OR Malnourish*:ab,ti OR 'Nutritional Deficienc*':ab,ti OR Undernutrition:ab,ti OR Undernourish*:ab,ti OR 'Nutritional Status':ab,ti OR 'Nutrition Status':ab,ti)

AND

('tuberculosis'/exp OR Tuberculosis:ab,ti OR TB:ab,ti OR tuberculoses:ab,ti) NOT ('animal'/exp NOT 'human'/exp) NOT 'child'/exp NOT 'review'/exp

Pubmed

("Malnutrition"[Mesh] OR Malnutrition [tiab] OR Malnourish* [tiab] OR Nutritional Deficienc* [tiab] OR Undernutrition [tiab] OR Undernourish* [tiab] OR "Nutritional Status"[Mesh] OR Nutritional Status [tiab] OR Nutrition Status [tiab])

AND

("Tuberculosis"[Mesh] OR Tuberculosis [tiab] OR TB [tiab] OR tuberculoses [tiab]) NOT ("animals"[MeSH] NOT "humans"[MeSH]) NOT "Child"[Mesh] NOT "Review" [Publication Type]

Cinahl

((MH "Malnutrition" OR MH "Nutritional Status" OR TI (Malnutrition OR Malnourish* OR Nutritional Deficienc* OR Undernutrition OR Undernourish* OR Nutritional Status OR Nutrition Status) OR AB (Malnutrition OR Malnourish* OR Nutritional Deficienc* OR Undernutrition OR Undernourish* OR Nutritional Status OR Nutrition Status)) AND ((MH "Tuberculosis+" OR TI (tuberculosis OR tb OR tuberculoses) OR AB (tuberculosis OR tb OR tuberculoses))

For peer review only

Supplement material 2. Details of included studies and risk of bias assessment

First author	Year of publication	Year(s) of data collection	Country	Rationale	Aim of malnutrition assessment	Assessment method	Clear cut-off	Maln* in results	Maln* in discussion	Risk of bias	Number of in/out patients	HIV coinfection	Susceptibility	Type of TB
Martins[24]	2009	2005-2006	Timor-Leste	+	Secondary, part of clinical outcome	BMI < 18.5 kg/m^2	-	+	+	medium	270 outpatients	unknown	unknown	pulmonary
Pakasi[25]	2009	unknown	Indonesia	++	Primary, prevalence and association	BMI < 18.5 kg/m^2	++	++	++	very low	121 outpatients	unknown	unknown	pulmonary
Ulasli[26]	2009	2001-2006	Turkey	+	Secondary, association	BMI < 20.0 kg/m^2	++	+	+	low	24 inpatients	unknown	unknown	pulmonary and/or extra pulmonary
Kim[27]	2010	2005-2006	South-Korea	++	Primary, malnutrition etiology	PIBW, BMI, Albumin, TLC, Cholesterol, Hb	++	++	++	very low	23, unknown type of patients	unknown	unknown	pulmonary
Khoharo[28]	2010	2007-2008	Pakistan	-	Secondary, risk factor	BMI < 18.5 kg/m^2	++	+	++	low	350, unknown type of patients	unknown	unknown	pulmonary
Pakasi[29]	2010	2004-2005	Indonesia	+	Secondary, part of clinical outcome	BMI (no clear cut-off)	-	+	+	medium	300 outpatients	unknown	unknown	pulmonary
Gambhir[30]	2010	2006-2009	India	-	Secondary, risk factor	BMI <18.5 kg/m^2	++	+	+	medium	95 inpatients	no	unknown	pulmonary and/or

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

														extra pulmonary
Singla[31]	2010	2004-2009	India	+	Secondary, risk factor	BMI < 18.5 kg/m^2	+	+	+	medium	175 in- and outpatients	no	unknown	pulmonary and/or extra pulmonary
Mupere[32]	2010	2002-2008	Uganda	++	Primary, association	BMI < 18.5 kg/m^2	+	+	+	low	445, unknown type of patients	115	unknown	pulmonary
Warmelink[33]	2010	2005-2008	Nederland	+	Primary, risk factor	Change in body weight	-	+	+	medium	192 inpatients	15	MDR patients included	pulmonary and/or extra pulmonary
Podewils[34]	2011	2000-2004	Latvia	++	Primary, association	BMI < 18.5 kg/m^2	++	++	+	very low	995 in- and outpatients	32	only MDR patients	pulmonary
De Jong[35]	2011	unknown	Gambia	-	Secondary, part of clinical outcome	BMI < 16 kg/m^2	++	+	+	medium	692, unknown type of patients	56	unknown	pulmonary
Kawai[36]	2011	2000-2005	Tanzania	+	Primary, follow-up of malnutrition	BMI < 18.5 kg/m^2	++	+	+	low	887 outpatients	471	unknown	pulmonary
Miyata[37]	2011	unknown	Japan	++	Primary, prognostic factor	SGA	++	++	++	very low	39 inpatients	unknown	unknown	pulmonary

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Mupere[38]	2012	unknown	Uganda	-	Primary, association	BMI < 18.5 kg/m^2	++	+	+	medium	747 outpatients	539	unknown	pulmonary
Piva[39]	2013	2008-2009	Brazil	++	Primary, prevalence	BMI < 18.5 kg/m^2	++	++	++	very low	34 in- and outpatients	unknown	unknown	pulmonary
Islam[40]	2013	2010-2011	Bangladesh	++	Primary, prevalence	BMI < 18.5 kg/m^2 and MUAC < 22 cm	++	+	++	very low	1068, unknown type of patients	unknown	unknown	pulmonary
Chittoor[41]	2013	unknown	Mexico	-	Secondary, association	Self-reported estimate of diet quality	-	+	+	high	75 outpatients	unknown	unknown	pulmonary
Miyata[42]	2013	unknown	Japan	+	Primary, prognostic factor	MNA < 17	++	+	+	low	53 inpatients	unknown	unknown	pulmonary
Bhargava[3]	2013	2004-2009	India	++	Primary, prevalence and association	BMI < 18.5 kg/m^2	++	++	++	very low	1695 in- and outpatients	39	unknown	pulmonary
Bakari[43]	2013	2009-2010	Tanzania	++	Primary, prevalence and follow-up of malnutrition	BMI < 18.5 kg/m^2	++	+	+	low	43 outpatients	43	unknown	pulmonary
Ismawati[44]	2013	2011-2012	Indonesia	+	Secondary, part of clinical outcome	BMI (no clear cut-off)	-	+	+	medium	30, unknown type of patients	unknown	unknown	pulmonary
Miyata[45]	2013	unknown	Japan	++	Primary, prognostic factor	MUST	++	+	+	low	57 inpatients	unknown	unknown	pulmonary

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Maeda[46]	2014	2007-2009	Vietnam	-	Secondary, association	BMI ≤ 18.5 kg/m^2	++	+	+	medium	465 outpatients	38	Including different types of resistance	pulmonary
Oliveira[47]	2014	2007-2010	Brazil	++	Secondary, association	TST AMA	+	++	++	very low	166 inpatients	31	unknown	pulmonary
Tian[48]	2014	2000-2001	China	-	Secondary, risk factor	BMI < 18.5 kg/m^2 and/or serum albumin < 30 g/L	++	+	+	medium	160 in- and outpatients	43	unknown	pulmonary
Kumar[49]	2014	2011-2014	India	+	Primary, prognostic factor	BMI (no clear cut-off)	-	+	+	medium	376 outpatients	unknown	Only MDR	pulmonary
Bacelo[50]	2015	2008-2013	Brazil	++	Primary, follow-up of malnutrition intervention	Multiple anthropometric and biochemical biomarkers	++	++	++	very low	68, unknown type of patients	22	unknown	Pulmonary and extra pulmonary
Golemba[51]	2015	2011-2014	Argentina	-	Secondary, association	BMI ≤ 20 kg/m^2	++	+	+	medium	75 inpatients	0	unknown	pulmonary
Te Brake[52]	2015	unknown	Indonesia	++	Primary, prognostic factor	BMI < 18.5 kg/m^2	++	+	++	very low	36 outpatients	0	unknown	pulmonary
Medellin-Garibay[53]	2015	unknown	Mexico	-	Secondary, association	BMI < 18.49 kg/m^2	+	+	+	medium	48, unknown type of patients	0	unknown	Pulmonary and extrapulmonary
Ezeamama[54]	2015	2004-2008	Uganda	++	Secondary, association	BMI < 18.5 kg/m^2	++	++	++	very low	208 outpatients	208	unknown	pulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

McLachlan[55]	2016	2014	South Africa	-	Secondary, association	BMI < 18.5 kg/m^2	++	-	+	medium	105 inpatients	74	Susceptible and MDR	Pulmonary and extrapulmonary
Gebrecherkos[56]	2016	2015	Ethiopia	+	Secondary, risk factor	BMI < 18.5 kg/m^2	++	+	+	Low	15 outpatients	4	No rifampicin resistance	pulmonary
Araújo-Mariz[57]	2016	2007-2012	Brazil	-	Secondary, prognostic factor	BMI < 18.5 kg/m^2	++	+	+	medium	173, unknown type of patients	173	unknown	unknown
Buntoro[58]	2016	unknown	Indonesia	+	Primary, part of clinical outcome	BMI <18.5 kg/m^2	++	++	++	very low	72 outpatients	unknown	unknown	pulmonary
Pandit[59]	2016	2012-2013	India	-	Secondary, prognostic factor	BMI <20.0 kg/m^2	-	+	+	high	148, unknown type of patients	unknown	unknown	pulmonary
Abdelbary[60]	2017	2006-2013	Mexico	-	Secondary, risk factor	Underweight	-	++	-	high	8431 outpatients	447	Susceptible and MDR	pulmonary and extrapulmonary
Bhat[61]	2017	2013	India	+	Secondary, association	BMI < 18.5 kg/m^2	++	+	+	low	267 outpatients	0	unknown	pulmonary
Hochberg[62]	2017	2014-2016	India	-	Primary, prevalence	BMI < 18.5 kg/m^2	++	++	++	low	409 outpatients	1	Only normal susceptible	pulmonary
Piparva[63]	2018	2014-2015	India	-	Secondary, part of clinical outcome	BMI < 18.5 kg/m^2	++	++	++	low	108 inpatients	5	only MDR	pulmonary

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Gurung[64]	2018	2016	Nepal	++	Primary, prevalence	BMI (no clear cut-off)	-	++	++	low	133 outpatients	1	unknown	pulmonary and extrapulmonary
Pande[65]	2018	2015-2016	India	-	Secondary, prevalence and association	BMI < 18.4 kg/m^2	+	+	+	medium	728 inpatients	53	unknown	pulmonary and extrapulmonary
Rao[66]	2018	2013-2014	India	-	Secondary, risk factor	BMI (no clear cut-off)	-	+	++	medium	220 outpatients	unknown	unknown	pulmonary
Sattler[67]	2018	unknown	4 continents, 26 study sites	+	Primary, association	BMI < 18.5 kg/m^2	-	+	+	medium	51, unknown type of patients	51	unknown	pulmonary
Kirchmann Lazzari[68]	2018	unknown	Brazil	++	Primary, prevalence	BMI TSF MAMC SGA	+	++	++	very low	108 inpatients	44	unknown	pulmonary
Cheng[69]	2019	2013-2016	China	-	Secondary, part of clinical outcome	BMI < 18 kg/m^2	+	+	-	high	85 inpatients	0	unknown	intestinal
Cavalheiro Skupien[70]	2019	unknown	Brazil	-	Secondary, association	BMI ≤ 18.5 kg/m^2	-	+	+	high	35 inpatients	unknown	unknown	pulmonary

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abdullahi[71]	2019	2012 - 2016	Kenya	+	Primary, association	BMI < 18.5 kg/m^2	++	++	-	low	10717 outpatients	3163	unknown	Pulmonary and extrapulmonary
Benzekri[72]	2019	2016 - 2017	Senegal	-	Primary, part of clinical outcome	BMI (no clear cut-off)	-	++	+	medium	26 outpatients	26	Susceptible	pulmonary
Chebrolu[73]	2019	unknown	USA	++	Primary, inclusion criterion	BMI < 16 kg/m^2	++	+	+	low	27, unknown type of patients	0	unknown	Pulmonary and extrapulmonary
Da Silva[74]	2019	2017 – 2018	Brazil	+	Primary, association	BMI < 18.5 kg/m^2 TSF MUAC MAMC BIA FFQ	+	++	++	low	35, unknown type of patients	14	unknown	Pulmonary
Feleke[75]	2019	2015 - 2018	Ethiopia	++	Primary, prevalence	BMI < 18.5 kg/m^2	++	++	+	very low	1681, unknown type of patients	595	unknown	Pulmonary and extrapulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Gashaw[76]	2019	2015 – 2017	Ethiopia	++	Primary, prevalence	BMI ≤ 18.5 kg/m^2 MUAC ≤ 23 cm (men) MUAC ≤ 22 cm (women)	++	++	++	very low	384, unknown type of patients	unknown	unknown	Pulmonary and extrapulmonary
Hoyt[77]	2019	2015 – 2017	India	++	Primary, part of clinical outcome	BMI < 18.5 kg/m^2	++	++	++	very low	173, unknown type of patients	unknown	unknown	pulmonary
Hussien[78]	2019	2017 – 2018	Ethiopia	++	Primary, prevalence	BMI < 18.5 kg/m^2	+	++	++	very low	372 inpatients	42	unknown	pulmonary
Lee[79]	2019	2016 – 2017	Philippines	++	Primary, prognostic factor	BMI< 17 kg/m^2 MUAC ≤ 20.5 cm (men) MUAC ≤ 18.5 cm (women)	++	++	++	very low	348 inpatients	22	31 MDR	unknown
Mailu[80]	2019	2013-2017	Kenya	-	Primary, prevalence	BMI < 18.5 kg/m^2	++	++	+	low	421409 outpatients	134776	unknown	Pulmonary and extrapulmonary

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Rashak[81]	2019	2010-2014	Mexico	++	Primary, prevalence and association	BMI < 18.5 kg/m^2 (or BMI ≤ 18.5 kg/m^2, not clearly described)	-	++	++	low	5508 , unknown type of patients	224	unknown	Pulmonary and extrapulmonary
Ren[82]	2019	2015 – 2017	China	++	Primary, prevalence	BMI < 18.5 kg/m^2	++	-	++	low	300 in- and outpatients	unknown	unknown	pulmonary
Wardani[83]	2019	2016	Indonesia	+	Secondary, risk factor	BMI ≤ 18.5 kg/m^2	++	+	-	medium	311 outpatients	unknown	unknown	pulmonary
Wessels[84]	2019	2015	South Africa	++	Secondary, association	BMI < 18.5 kg/m^2 (underweight)	-	++	+	low	100 inpatients	68	unknown	unknown
White[85]	2019	2016 – 2017	Philippines	++	Primary and secondary, validation of tool and prognostic factor	BMI < 17 kg/m^2	++	++	++	very low	348 inpatients	22	31	unknown

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Ma'rufi[86]	2020	2017	Indonesia	++	Primary, follow-up of malnutrition	BMI < 18.5 kg/m^2	++	++	++	very low	200 outpatients	0 (excluded)	0 (excluded)	pulmonary
Musuenge[87]	2020	unknown	Burkino Faso	++	Primary, prevalence and association	BMI < 18.5 kg/m^2	++	++	++	Very low	302 outpatients	23	unknown	pulmonary
Seid[88]	2020	2019	Ethiopia	++	Primary, prevalence and prognostic factor	BMI < 18.5 kg/m^2	+	++	++	Very low	284 in- and outpatients	51	Unknown	Pulmonary and extrapulmonary
Edwards[89]	2020	2017	Phillipines	+	Primary, prevalence	BMI < 17.0 kg/m^2	+	++	++	Very low	446 outpatients	68 (28%)	Susceptible	Unknown
White[90]	2020	2017	Phillipines	++	Primary, prevalence and association	BMI < 18.5 kg/m^2	+	++	++	Very low	637 outpatients	74 (24%)	Unknown	Unknown

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Mollah[91]	2020	2018	India	++	Primary, prevalence	BMI < 18.5 kg/m^2 MUAC <23 (m) & <22 (f)	++	-	++	Low	113, unknown type of patients	Unknown	Unknown	Pulmonary and extrapulmonary

Abbreviations *PIBW: Percent Ideal Body Weight; BMI: Body Mass index; TLC: Total Lymphocyte Count; Hb: Haemoglobin; MUAC: Mid Upper Arm Circumference; MNA: Mini Nutritional Assessment; AMA: Arm Muscle Area; MAMC: Mid Arm Muscle Circumference; TSF: Triceps Skin Fold ; TST: Triceps Skinfold Thickness; SGA: Subjective Global Assessment; BIA: Bio-electrical Impedance Analysis; FFQ: Food Frequency Questionnaire*

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

3 Bhargava A, Chatterjee M, Jain Y, et al. Nutritional status of adult patients with pulmonary tuberculosis in rural central India and its association with mortality. *PLoS One* 2013;8:e77979 doi:10.1371/journal.pone.0077979 [doi].

24 Martins N, Morris P, Kelly PM. Food incentives to improve completion of tuberculosis treatment: randomised controlled trial in Dili, Timor-Leste. *BMJ* 2009;339:b4248 doi:10.1136/bmj.b4248 [doi].

25 Pakasi TA, Karyadi E, Dolmans WM, et al. Malnutrition and socio-demographic factors associated with pulmonary tuberculosis in Timor and Rote Islands, Indonesia. *Int J Tuberc Lung Dis* 2009;13:755-9.

26 Ulasli SS, Ulubay G, Arslan NG, et al. Characteristics and Outcomes of End-Stage Renal Disease Patients With Active Tuberculosis Followed in Intensive Care Units. *Saudi J Kidney Dis Transpl* 2009;20:254-9.

27 Kim JS, Wilson JM, Lee SR. Dietary implications on mechanisms of sarcopenia: roles of protein, amino acids and antioxidants. *J Nutr Biochem* 2010;21:1-13 doi:10.1016/j.jnutbio.2009.06.014; 10.1016/j.jnutbio.2009.06.014.

28 Khoharo HK, Ansari S, Siddiqui AA, et al. Standard antituberculosis drug induced hepatotoxicity: Do the risk factors matter?. *J Liaquat Uni Med Health Sci* 2010;9:84-7.

29 Pakasi TA, Karyadi E, Suratih NMD, et al. Zinc and vitamin A supplementation fails to reduce sputum conversion time in severely malnourished pulmonary tuberculosis patients in Indonesia. *Nutr J* 2010;9:41.

30 Gambhir HS, Kaushik RM, Kaushik R, et al. Tobacco smoking-associated risk for tuberculosis: a case-control study. *Int Health* 2010;2:216-22.

31 Singla R, Sharma SK, Mohan A, et al. Evaluation of risk factors for antituberculosis treatment induced hepatotoxicity. *Indian J Med Res* 2010;132:81-6.

32 Mupere E, Zalwango S, Chiunda A, et al. Body composition among HIV-seropositive and HIV-seronegative adult patients with pulmonary tuberculosis in Uganda. *Ann Epidemiol* 2010;20:210-6.

33 Warmelink I, van Altena R, ten Hacken N, et al. Nutritional status and vitamin D3 during antimicrobial treatment. *Lancet* 2011;377:1407-8.

34 Podewils LJ, Holtz T, Riekstina V, et al. Impact of malnutrition on clinical presentation, clinical course, and mortality in MDR-TB patients. *Epidemiol Infect* 2011;139:113-20 doi:10.1017/S0950268810000907 [doi].

35 de Jong BC, Adetifa I, Walther B, et al. Differences between tuberculosis cases infected with Mycobacterium africanum, West African type 2, relative to Euro-American Mycobacterium tuberculosis: an update. *FEMS Immunol Med Microbiol* 2010;58:102-5 doi:10.1111/j.1574-695X.2009.00628.x [doi].

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

36 Kawai K, Villamor E, Mugusi FM, et al. Predictors of change in nutritional and hemoglobin status among adults treated for tuberculosis in Tanzania. *Int J Tuberc Lung Dis* 2011;15:1380-9 doi:10.5588/ijtld.10.0784 [doi].

37 Miyata S, Tanaka M, Ihaku D. Subjective global assessment in patients with pulmonary tuberculosis. *Nutr Clin Pract* 2011;26:55-60 doi:10.1177/0884533610392380 [doi].

38 Mupere E, Malone L, Zalwango S, et al. Lean tissue mass wasting is associated with increased risk of mortality among women with pulmonary tuberculosis in urban Uganda. *Ann Epidemiol* 2012;22:466-73.

39 Piva SG, Costa Mda C, Barreto FR, et al. Prevalence of nutritional deficiency in patients with pulmonary tuberculosis. *J Bras Pneumol* 2013;39:476-83 doi:10.1590/S1806-37132013000400012 [doi].

40 Islam QS, Ahmed SM, Islam MA, et al. Beyond drugs: tuberculosis patients in Bangladesh need nutritional support during convalescence. *Public health action* 2013;3:136-40.

41 Chittoor G, Arya R, Farook VS, et al. Epidemiologic investigation of tuberculosis in a Mexican population from Chihuahua State, Mexico: a pilot study. *Tuberculosis (Edinburgh, Scotland)* 2013;93:S71-7.

42 Miyata S, Tanaka M, Ihaku D. Full mini nutritional assessment and prognosis in elderly patients with pulmonary tuberculosis. *J Am Coll Nutr* 2013;32:307-11 doi:10.1080/07315724.2013.826114 [doi].

43 Bakari M, Wamsele J, MacKenzie T, et al. Nutritional status of HIV-infected women with tuberculosis in Dar es Salaam, Tanzania. *Public health action* 2013;3:224-9.

44 Ismawati. R., Bambang W, Priatna DY, et al. The effect of zinc, lysine and vitamin a supplementation to increase cellular immune response of pulmonary tuberculosis patients. *J Trace Elem Med Biol* 2013;27:21.

45 Miyata S, Tanaka M, Ihaku D. The prognostic significance of nutritional status using malnutrition universal screening tool in patients with pulmonary tuberculosis. *Nutrition journal* 2013;12:42.

46 Maeda S, Hang NT, Lien LT, et al. Mycobacterium tuberculosis strains spreading in Hanoi, Vietnam: Beijing sublineages, genotypes, drug susceptibility patterns, and host factors. *Tuberculosis (Edinb)* 2014;94:649-56 doi:S1472-9792(14)20533-6 [pii].

47 Oliveira MG, Delogo KN, Oliveira,Hedi Marinho de Melo Gomes de, et al. Anemia in hospitalized patients with pulmonary tuberculosis. *Jornal brasileiro de pneumologia : publicacao oficial da Sociedade Brasileira de Pneumologia e Tisilogia* 2014;40:403-10.

48 Tian PW, Wang Y, Shen YC, et al. Different risk factors of recurrent pulmonary tuberculosis between Tibetan and Han populations in Southwest China. *Eur Rev Med Pharmacol Sci* 2014;18:1482-6 doi:7406 [pii].

49 Kumar NSS, Hemraj SK, Dutt RA. Phase angle measurement in pulmonary tuberculosis patients and control subjects using bio-impedance analysis. *The Indian journal of tuberculosis* 2014;61:224-31.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

50 Bacelo AC, Ramalho A, Brasil PE, et al. Nutritional Supplementation Is a Necessary Complement to Dietary Counseling among Tuberculosis and Tuberculosis-HIV Patients. *PLoS One* 2015;10:e0134785 doi:10.1371/journal.pone.0134785 [doi].

51 Golemba AS, Ferreyra FG, Martearena RE, et al. Drug-induced hepatotoxicity and tuberculosis in a hospital from the Argentinian northeast: cross-sectional study. *Medwave* 2015;15:e6135 doi:10.5867/medwave.2015.04.6135 [doi].

52 Te Brake LH, Ruslami R, Later-Nijland H, et al. Exposure to total and protein-unbound rifampicin is not affected by malnutrition in Indonesian tuberculosis patients. *Antimicrob Agents Chemother* 2015;74:986-90 doi:AAC.03485-14 [pii].

53 Medellin-Garibay SE, Cortez-Espinosa N, Milan-Segovia RC, et al. Clinical Pharmacokinetics of Rifampin in Patients with Tuberculosis and Type 2 Diabetes Mellitus: Association with Biochemical and Immunological Parameters. *Antimicrob Agents Chemother* 2015;59:7707-14 doi:10.1128/AAC.01067-15 [doi].

54 Ezeamama AE, Mupere E, Oloya J, et al. Age, sex, and nutritional status modify the CD4+ T-cell recovery rate in HIV-tuberculosis co-infected patients on combination antiretroviral therapy. *Int J Infect Dis* 2015;35:73-9 doi:10.1016/j.ijid.2015.04.008 [doi].

55 McLachlan I, Visser WI, Jordaan HF. Skin conditions in a South African tuberculosis hospital: Prevalence, description, and possible associations. *Int J Dermatol* 2016;55:1234-41.

56 Gebrecherkos T, Gelaw B, Tessema B. Smear positive pulmonary tuberculosis and HIV co-infection in prison settings of North Gondar Zone, Northwest Ethiopia. *BMC Public Health* 2016;16:1091.

57 Araújo-Mariz C, Lopes EP, Acioli-Santos B, et al. Hepatotoxicity during Treatment for Tuberculosis in People Living with HIV/AIDS. *PloS one* 2016;11:e0157725.

58 Buntoro IF, Kristin E, Sumardi. Decrease of liver function after treatment of antituberculosis drugs in tuberculosis patients with malnutrition and alcohol consumption. *Int J Pharm Pharm Sci* 2016;8:269-73.

59 Pandit A, Pandey Ak. Liver dysfunction in pulmonary tuberculosis patients on dots: A study and review. *J Gastroenterol Hepatol Res* 2016;5:2254-60.

60 Abdelbary BE, Garcia-Viveros M, Ramirez-Oropesa H, et al. Predicting treatment failure, death and drug resistance using a computed risk score among newly diagnosed TB patients in Tamaulipas, Mexico. *Epidemiol Infect* 2017;145:3020-34.

61 Bhat J, Rao VG, Sharma RK, et al. Investigation of the risk factors for pulmonary tuberculosis: A case-control study among Saharia tribe in Gwalior district, Madhya Pradesh, India. *Indian J Med Res* 2017;146:97-104.

62 Hochberg NS, Sarkar S, Horsburgh CR, et al. Comorbidities in pulmonary tuberculosis cases in Puducherry and Tamil Nadu, India: Opportunities for intervention. *PloS one* 2017;12:e0183195.

63 Piparva KG, Jansari G, Singh AP. Evaluation of treatment outcome and adverse drug reaction of directly observed treatment (DOT) plus regimen in multidrug-resistant tuberculosis (MDR-TB) patients at district tuberculosis centre Rajkot. *Perspectives in clinical research* 2018;9:165-9.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

64 Gurung LM, Bhatt LD, Karmacharya I, et al. Dietary Practice and Nutritional Status of Tuberculosis Patients in Pokhara: A Cross Sectional Study. *Frontiers in nutrition* 2018;5:1-6.

65 Pande T, Huddart S, Xavier W, et al. Prevalence of diabetes mellitus amongst hospitalized tuberculosis patients at an Indian tertiary care center: A descriptive analysis. *PloS one* 2018;13:e0200838.

66 Rao VG, Bhat J, Yadav R, et al. A comparative study of the socio-economic risk factors for pulmonary tuberculosis in the Saharia tribe of Madhya Pradesh, India. *Trans R Soc Trop Med Hyg* 2018;112:272-8.

67 Sattler FR, Chelliah D, Wu X, et al. Biomarkers Associated with Death After Initiating Treatment for Tuberculosis and HIV in Patients with Very Low CD4 Cells. *Pathogens & immunity* 2018;3:46-62.

68 Lazzari T.K., Forte G.C., Silva D.R. Nutrition Status Among HIV-Positive and HIV-Negative Inpatients with Pulmonary Tuberculosis. *Nutr.Clin.Prac.* 2018;33:858-64.

69 Cheng W, Zhang S, Li Y, et al. Intestinal tuberculosis: clinico-pathological profile and the importance of a high degree of suspicion. *Trop Med Int Health* 2019;24:81-90.

70 Skupien EC, Lazzari T, Coutinho SE, et al. The relation between leptin and inflammatory markers with respiratory and peripheral muscle strength in tuberculosis: A case-control study. *Clin Respir J* 2018;12:2559-65.

71 Abdullahi OA, Ngari MM, Sanga D, et al. Mortality during treatment for tuberculosis; a review of surveillance data in a rural county in Kenya. *PloS one* 2019;14:e0219191.

72 Benzekri NA, Sambou JF, Tamba IT, et al. Nutrition support for HIV-TB co-infected adults in Senegal, West Africa: A randomized pilot implementation study. *PloS one* 2019;14:e0219118.

73 Chebrolu P., Laux T., Chowdhury S., et al. The risk of refeeding syndrome among severely malnourished tuberculosis patients in Chhattisgarh, India. *Indian J Tuberc* 2019:https://doi.org/10.1016/j.ijtb.2019.03.004.

74 da Silva LF, Skupien EC, Lazzari T, et al. Advanced glycation end products (AGE) and receptor for AGE (RAGE) in patients with active tuberculosis, and their relationship between food intake and nutritional status. *PloS one* 2019;14:e0213991.

75 Feleke BE, Feleke TE, Biadglegne F. Nutritional status of tuberculosis patients, a comparative cross-sectional study. *BMC Pulm Med* 2019;19:182.

76 Gashaw F, Bekele S, Mekonnen Y, et al. High helminthic co-infection in tuberculosis patients with undernutritional status in northeastern Ethiopia. *Infect Dis Poverty* 2019;8:88.

77 Hoyt KJ, Sarkar S, White L, et al. Effect of malnutrition on radiographic findings and mycobacterial burden in pulmonary tuberculosis. *PloS one* 2019;14:e0214011.

78 Hussien B., Hussen M.M., Seid A., et al. Nutritional deficiency and associated factors among new pulmonary tuberculosis patients of Bale Zone Hospitals, southeast Ethiopia. *BMC Res Notes* 2019;12:751.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

79 Lee N, White LV, Marin FP, et al. Mid-upper arm circumference predicts death in adult patients admitted to a TB ward in the Philippines: A prospective cohort study. *PloS one* 2019;14:e0218193.

80 Mailu EW, Owiti P, Ade S, et al. Tuberculosis control activities in the private and public health sectors of Kenya from 2013 to 2017: how do they compare?. *Trans R Soc Trop Med Hyg* 2019;113:740-8.

81 Rashak HA, Sánchez-PÃ©rez HJ, Abdelbary BE, et al. Diabetes, undernutrition, migration and indigenous communities: tuberculosis in Chiapas, Mexico. *Epidemiol Infect* 2019;147:e71.

82 Ren Z, Zhao F, Chen H, et al. Nutritional intakes and associated factors among tuberculosis patients: a cross-sectional study in China. *BMC Infect Dis* 2019;19:907.

83 Wardani DWSR, Wahono EP. Predominant Determinants of Delayed Tuberculosis Sputum Conversion in Indonesia. *Indian J Commun Med* 2019;44:53-7.

84 Wessels J, Walsh CM, Nel M. Smoking habits and alcohol use of patients with tuberculosis at Standerton Tuberculosis Specialised Hospital, Mpumalanga, South Africa. *Health SA Gesondheid* 2019;24:1146.

85 White LV, Edwards T, Lee N, et al. Co-morbidities in filipino persons with tuberculosis: A cross-sectional study in urban and rural public TBDOTS facilities. *Trans R Soc Trop Med Hyg* 2019;113:S210.

86 Ma'rufi I, Ali K, Jati SK, et al. Improvement of Nutritional Status among Tuberculosis Patients by Channa striata Supplementation: A True Experimental Study in Indonesia. *BioMed Research International* 2020:1-10.

87 Musuenge BB, Poda GG, Chen PC. Nutritional Status of Patients with Tuberculosis and Associated Factors in the Health Centre Region of Burkina Faso. *Nutrients* 2020;12:10.3390/nu12092540 doi:E2540 [pii].

88 Seid G, Ayele M. Undernutrition and Mortality among Adult Tuberculosis Patients in Addis Ababa, Ethiopia. *Adv Prev Med* 2020;2020:5238010 doi:10.1155/2020/5238010 [doi].

89 Edwards T, White LV, Lee N, et al. Effects of comorbidities on quality of life in Filipino people with tuberculosis. *The international journal of tuberculosis and lung disease : the official journal of the International Union against Tuberculosis and Lung Disease* 2020;24:712-9.

90 White LV, Edwards T, Lee N, et al. Patterns and predictors of co-morbidities in Tuberculosis: A cross-sectional study in the Philippines. *Sci Rep* 2020;10:4100.

91 Mollah A, Shrivastava P, Das DK, et al. Nutritional status of adult Tuberculosis patients in Burdwan municipality area of West Bengal. *Indian Journal of Community Health* 2020;32:438-43.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplement Material 3: Overview of validated malnutrition screening tools available in English language[1]

Screening- tool	Characteristics	Target group
MUST[2]	BMI, weight loss and acute disease effect	All adult patients
PG-SGA Short Form[3]	Weight loss, food intake, nutrition impact symptoms, physical activity	All adult patients
SNAQ[4]	Weight loss, appetite, use of supplements/tube feeding	Adult hospitalized patients
MNA-SF[5]	Food intake, weight loss, mobility, psychological stress/acute disease, dementia or depression, BMI	Elderly patients
SNAQ 65+ [6]	Weight loss, MUAC, appetite, functionality	Elderly patients and/or rehabilitation care
SNAQ RC[7]	Weight loss, dependence, appetite, BMI	Elderly patients in residential care
Nutric Score[8]	Age, comorbidity, days from hospital to ICU admittance, APACHE score, SOFA score	Adult intensive care

References

1 Multidiciplinary Guidelines Malnutrition 2019. Available at: https://www.stuurgroepondervoeding.nl/wp-. Accessed june/19, 2021.

2 Elia M. The 'MUST 'report. Nutritional screening for adults: a multidisciplinary responsibility. Development and use of the 'Malnutrition Universal Screening Tool' (MUST) for adults. Worcs, United Kingdom: British Association for Parenteral and Enteral Nutrition (BAPEN) 2003.

3 Abbott J, Teleni L, McKavanagh D, et al. Patient-Generated Subjective Global Assessment Short Form (PG-SGA SF) is a valid screening tool in chemotherapy outpatients. *Support Care Cancer* 2016;24:3883-7 doi:10.1007/s00520-016-3196-0 [doi].

4 Kruizenga HM, Seidell JC, de Vet HC, et al. Development and validation of a hospital screening tool for malnutrition: the short nutritional assessment questionnaire (SNAQ). *Clin Nutr* 2005;24:75-82 doi:S0261-5614(04)00124-4 [pii].

5 Rubenstein LZ, Harker JO, Salva A, et al. Screening for undernutrition in geriatric practice: developing the short-form mini-nutritional assessment (MNA-SF). *J Gerontol A Biol Sci Med Sci* 2001;56:M366-72 doi:10.1093/gerona/56.6.m366 [doi].

For peer review only

6 Wijnhoven HA, Schilp J, van Bokhorst-de van der Schueren,M.A., et al. Development and validation of criteria for determining undernutrition in community-dwelling older men and women: The Short Nutritional Assessment Questionnaire 65+. *Clin Nutr* 2012;31:351-8 doi:10.1016/j.clnu.2011.10.013 [doi].

7 Kruizenga HM, de Vet HC, Van Marissing CM, et al. The SNAQ(RC), an easy traffic light system as a first step in the recognition of undernutrition in residential care. *J Nutr Health Aging* 2010;14:83-9.

8 Heyland DK, Dhaliwal R, Jiang X, et al. Identifying critically ill patients who benefit the most from nutrition therapy: the development and initial validation of a novel risk assessment tool. *Crit Care* 2011;15:R268 doi:10.1186/cc10546 [doi].

For peer review only



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Line 1, Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Line 47, Page 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Line 110-119, page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Line 120-122, page 5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Line 136-148, page 5
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Line 130-133, page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary material 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Line 151-156, page 5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Line 162-165, page 6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Line 172-180, page 6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Line 168-171, page 6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Line 187-199, page 6,7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Line 151-156, page 5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	NA
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1: Flow diagram
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	Supplementary material 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary material 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	NA
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Line 219-257
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Line 240-255, page 10
	23b	Discuss any limitations of the evidence included in the review.	Line 256-265, page 10
	23c	Discuss any limitations of the review processes used.	Line 266-274, page 10
	23d	Discuss implications of the results for practice, policy, and future research.	Line 283-288, page 10
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Line 70 (page 3) and 125-126 (page 5)
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Line 293-294, page 11
Competing interests	26	Declare any competing interests of review authors.	Line 297-299, page 11
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only